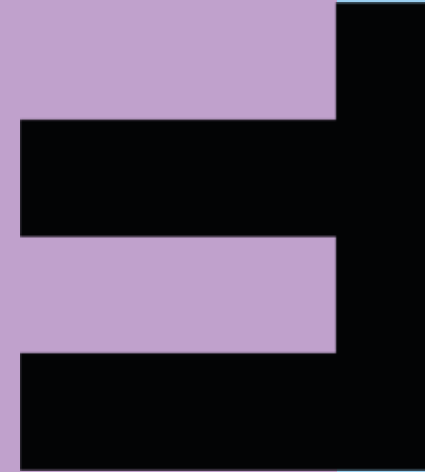


FHV

Vorarlberg University
of Applied Sciences



Technical Information Systems

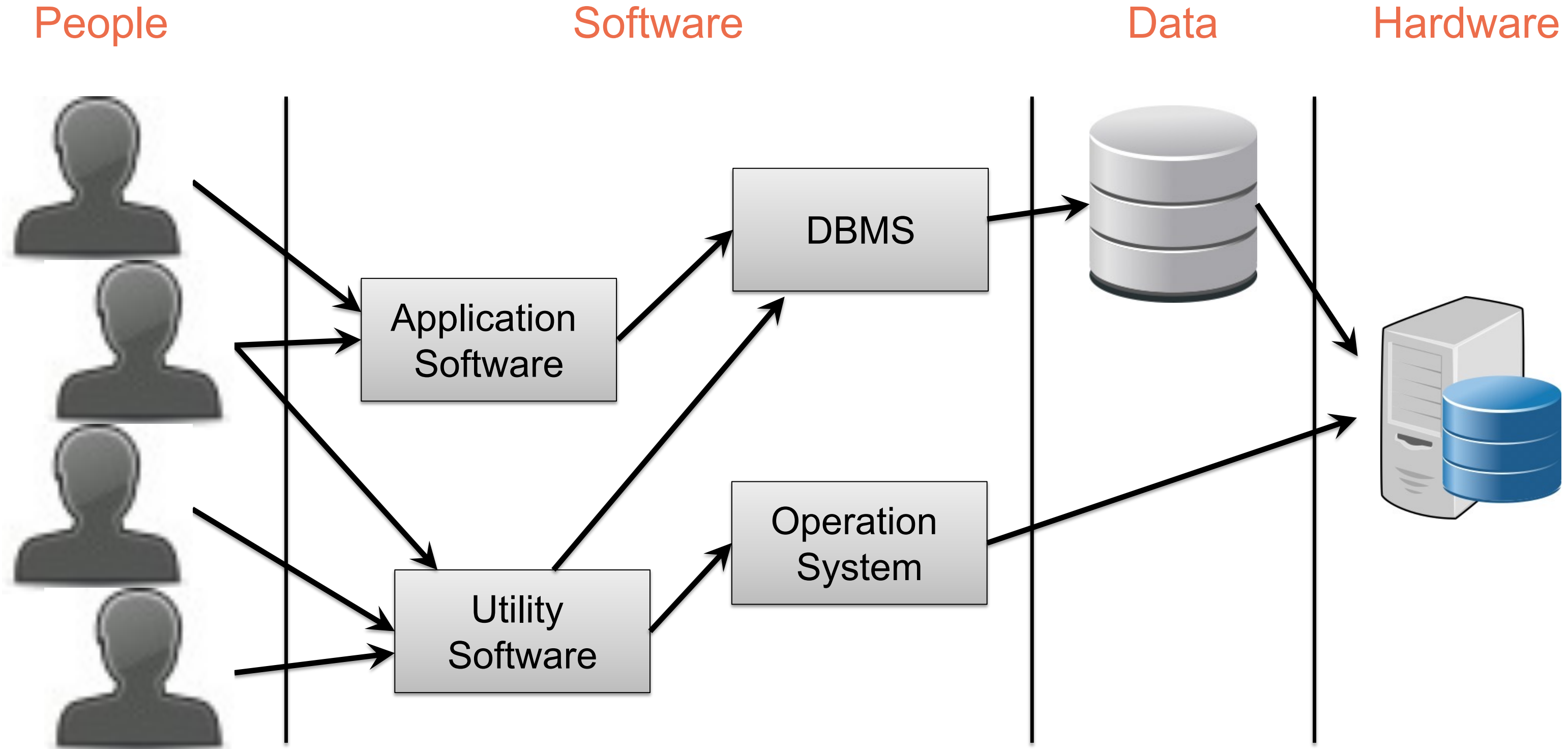
5 System Architecture
Data Quality
Transactions

5. Architecture, Data Quality, Transactions

1. Basic System Structure

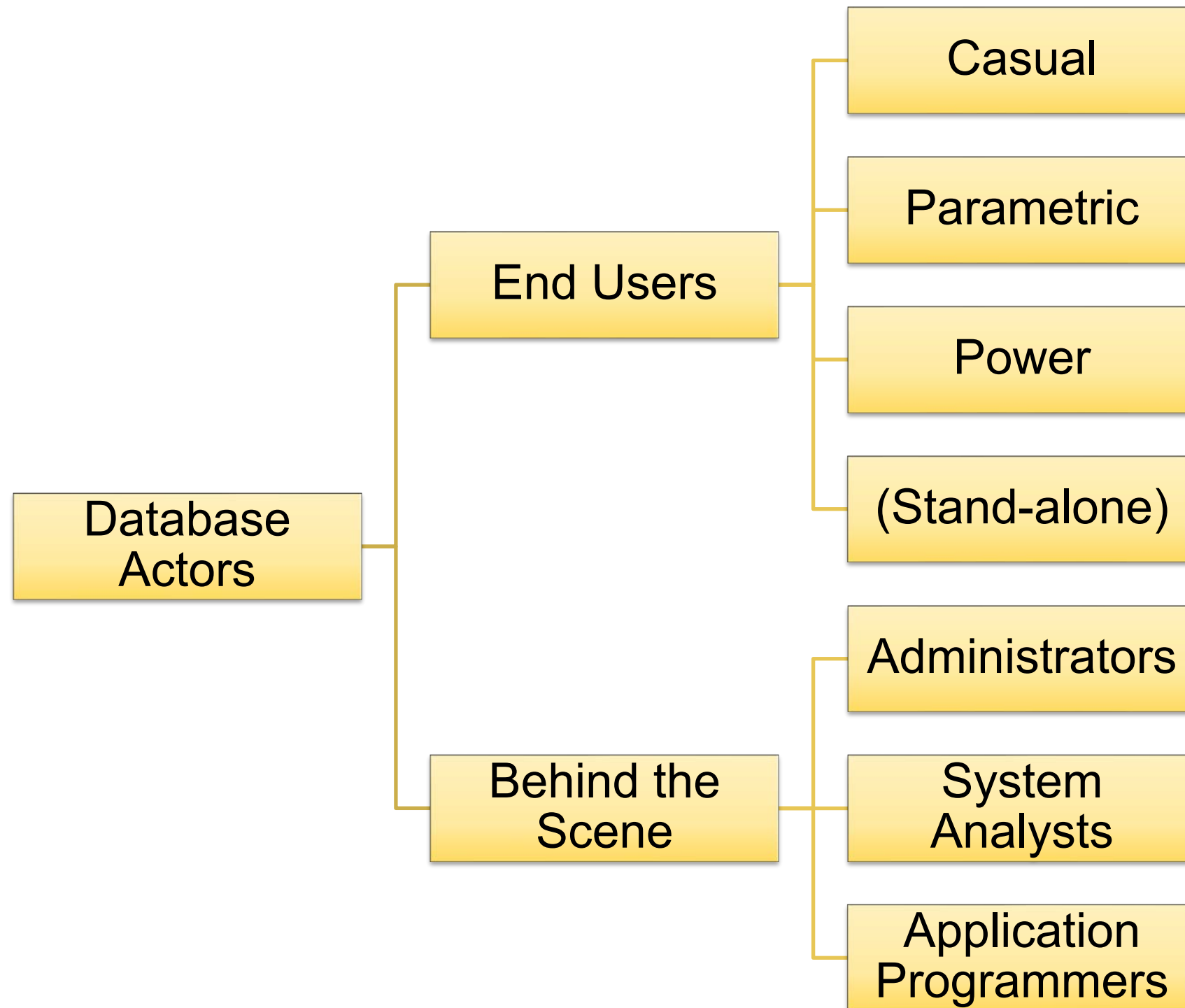
The Database System

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The Database System - Actors

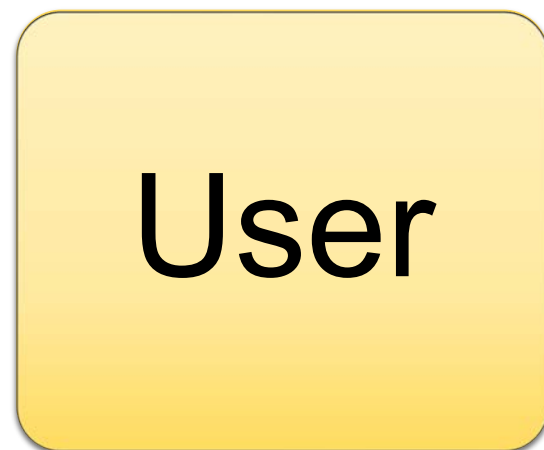
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Database

20 November 2023

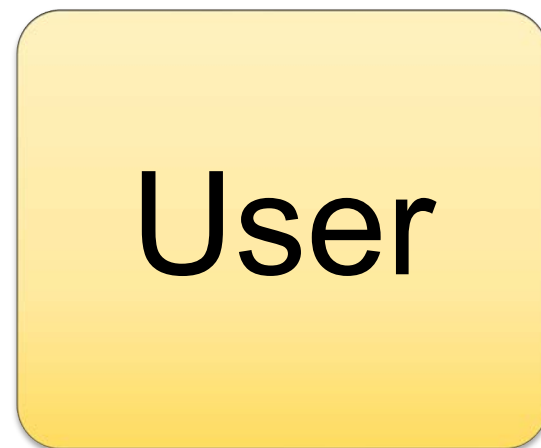
- A **database** represents some aspect of the real world, often called miniworld or universe of discourse (UoD), is a logically coherent collection of data with some inherent meaning and is designed, built, and populated with data for a specific purpose.



Database Management System

20 November 2023

- A **database management system (DBMS)** is a computer software system that interacts with the user, other applications, and the database itself to capture and analyze data. A general-purpose DBMS is designed to allow the definition, creation, querying, update, and administration of databases.



Database System

20 November 2023

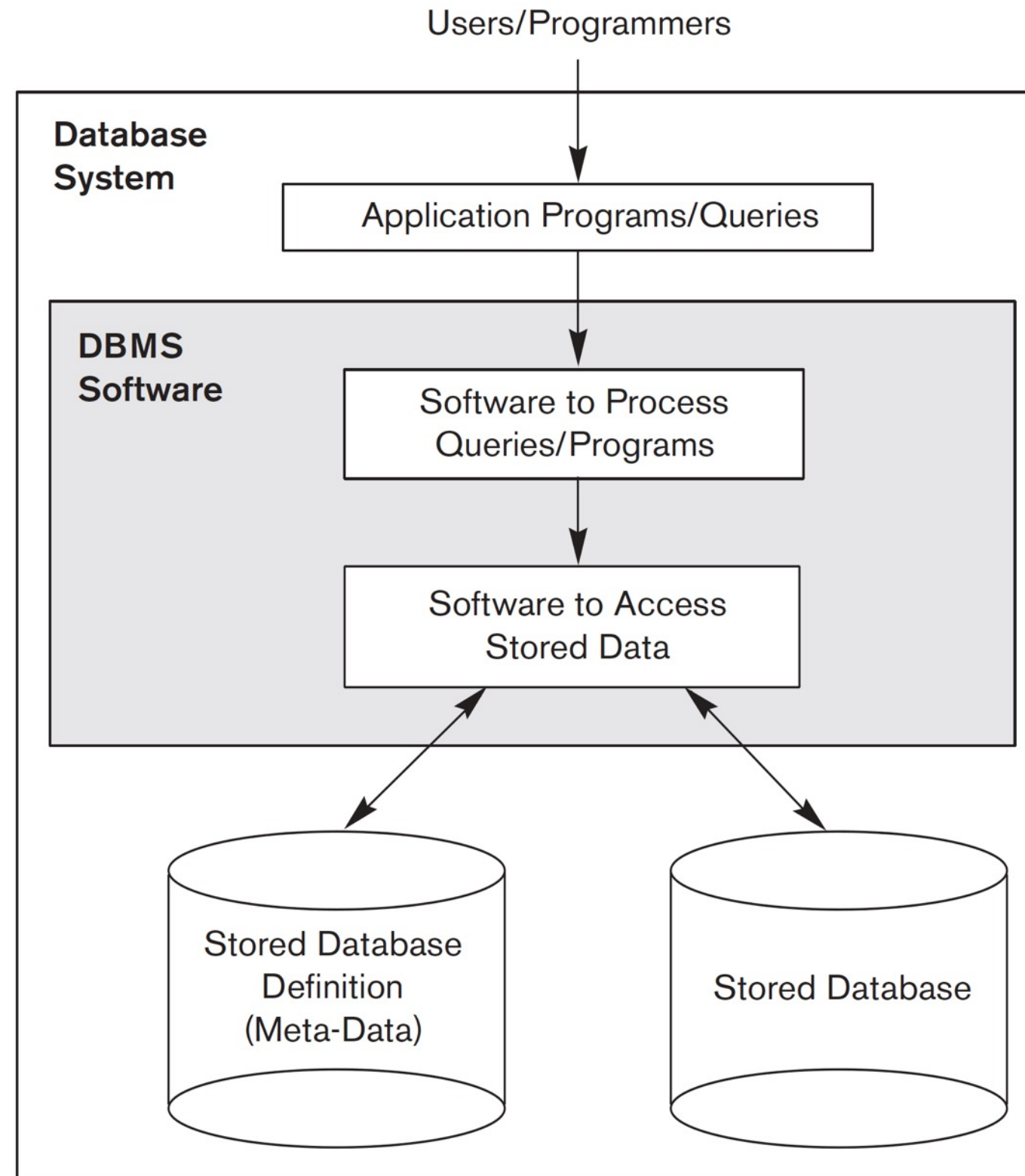
- Database System (DBS) = Database Management System
+ Database
+ Interface



Database System

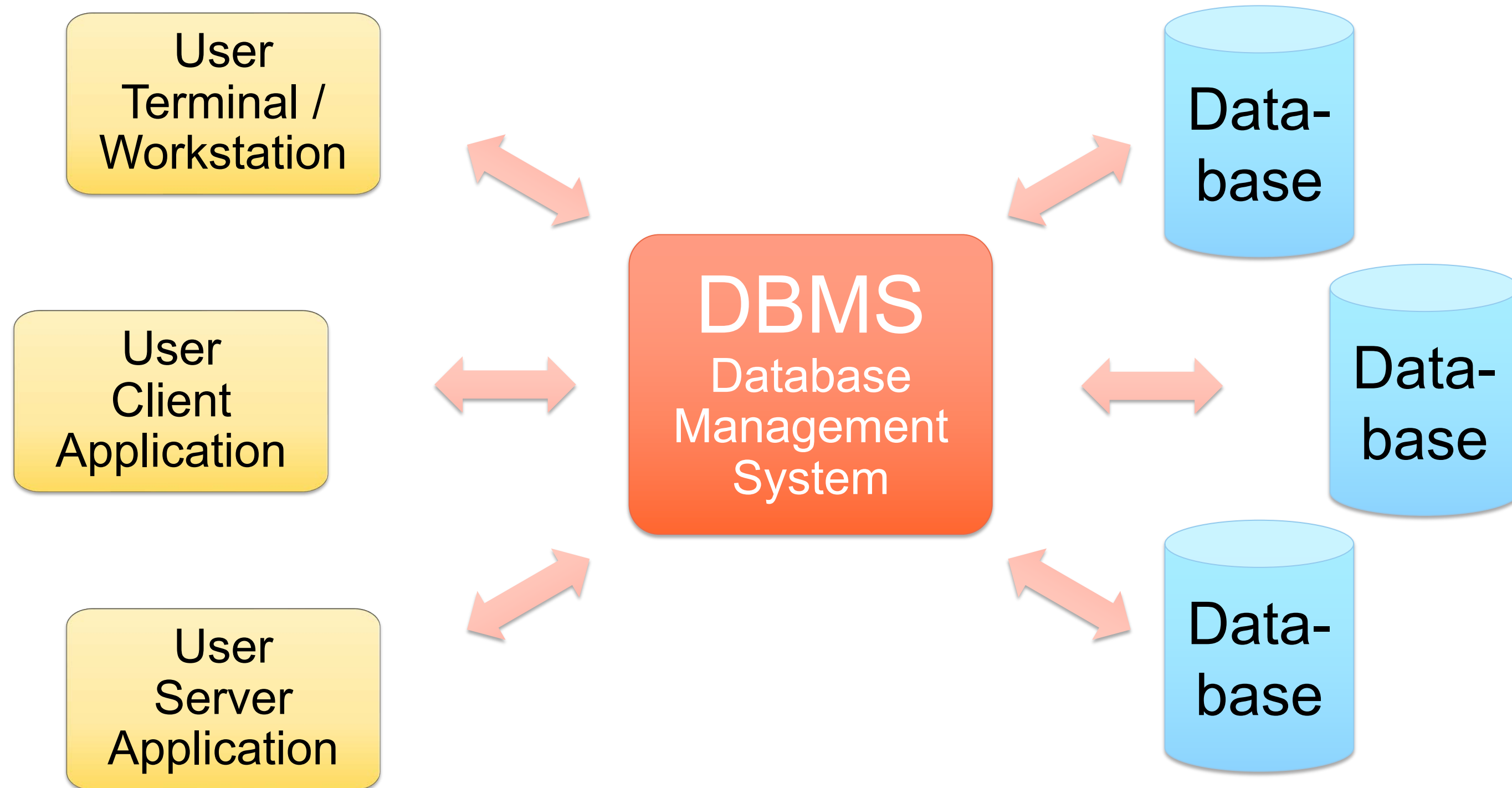
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Simplified
representation



Distributed Multiuser System

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Main Functions of a DBMS

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Definition

- Specifying the data types, structures, and constraints in the databases

Construction

- Storing the data itself on some storage medium

Manipulation

- Querying the database to retrieve specific data
- Updating the database to reflect changes
- Generating reports from the data

Sharing

- Controlled allowance for multiple users and programs to access the database (includes data protection)

System protection

- System protection against software- and hardware malfunctions
- Data Security against unauthorized or malicious access

Data protection, security, privacy

20 November 2023

- **Data protection** is the process of safeguarding important information from corruption, compromise or loss. Data protection centers around backup and recovery, although there are any number of data protection tools available
- **Data security** is the defense of digital information against internal and external, malicious and accidental threats. Although data security focuses specifically on keeping data secure, it also incorporates infrastructure security
- **Data privacy**, also called information privacy, is when an organization or individual must determine what data in a computer system can be shared with third parties.

There are two main aspects to data privacy. The first is **access control**. A big part of ensuring data privacy is determining who should have authorized access to the data and who shouldn't. The second aspect of data privacy involves putting mechanisms into place that will **prevent unauthorized access** to the data.

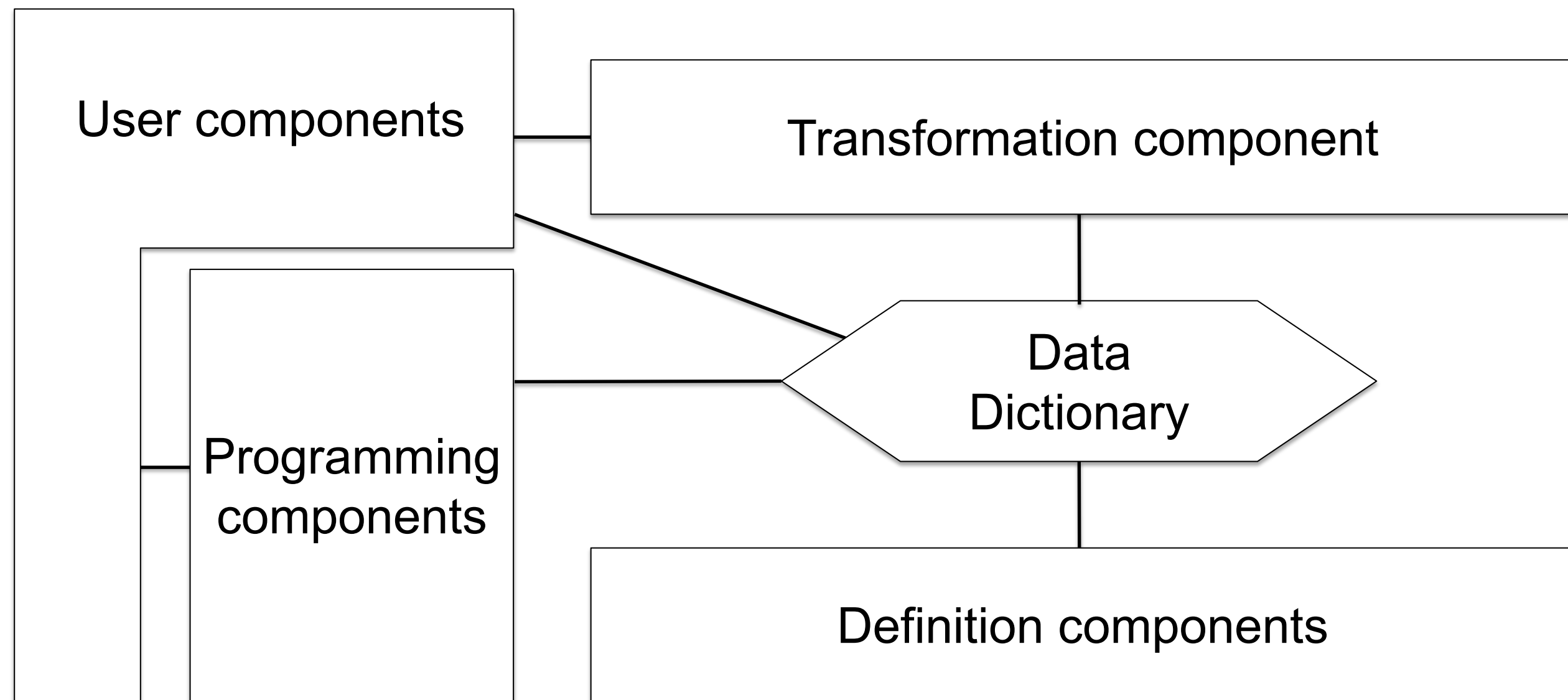
5. Architecture, Data Quality, Transactions

1. Basic System Structure
2. ANSI-SPARC Architecture

Database System architecture – Function Modules

ANSI-SPARC Architecture

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Database System architecture – Function Modules

ANSI-SPARC Architecture

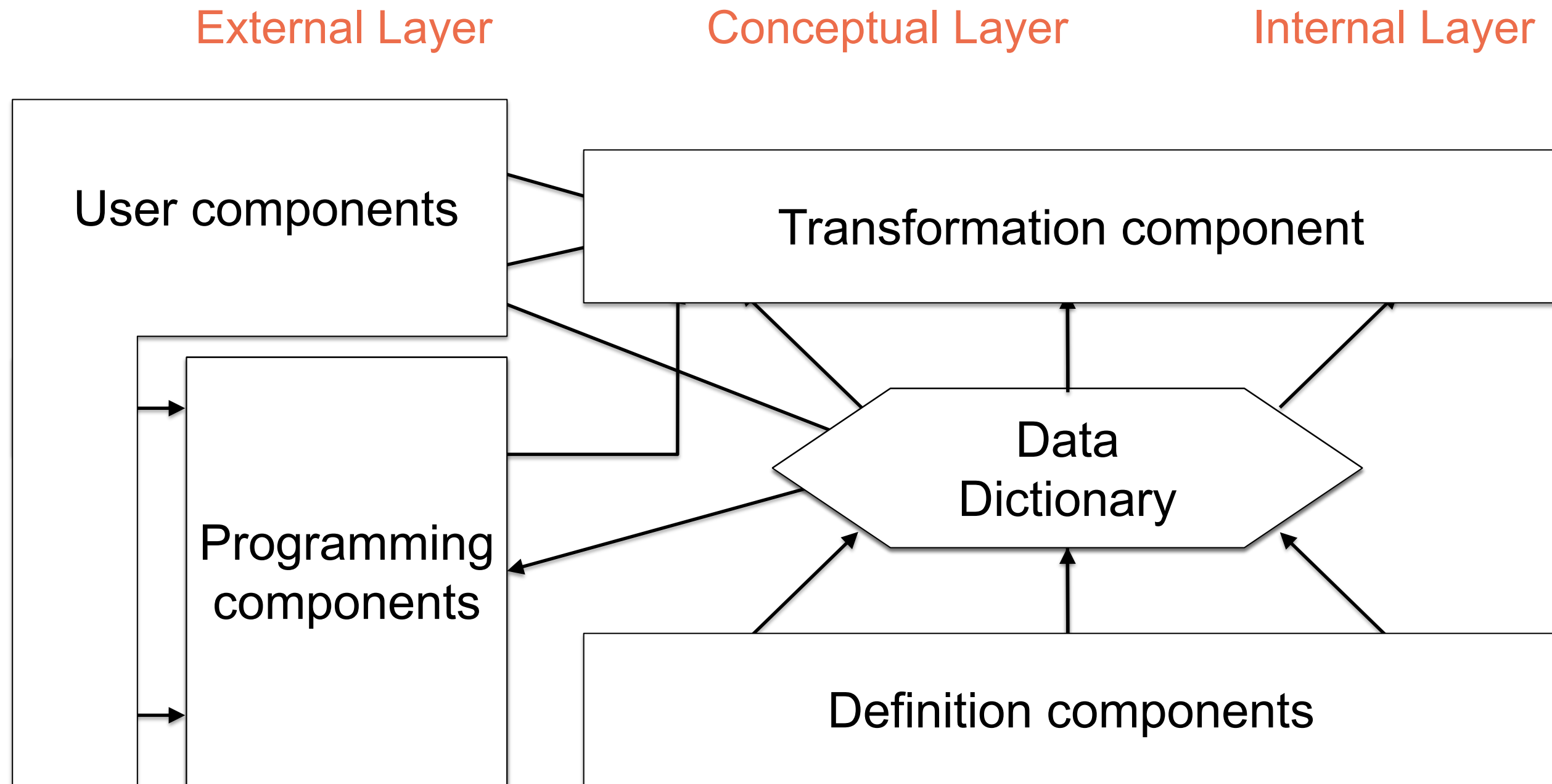
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- The **data dictionary** (or schema catalog) is the central element of the structure holding all the relevant information for database management.
- The **definition components** allow administrators to define data structure, data organization and access and what users can see
- **Programming components** contain the development environment to create an application with data processes and interface design
- User components contain direct access possibilities with different skill levels (created by the administrators or predelivered by the system)
- **Transformation components** convert select and change requests incrementally into disc accesses. Backwards Bytes are transferred to tables for the user view

Database System architecture – Function Modules

ANSI-SPARC Architecture

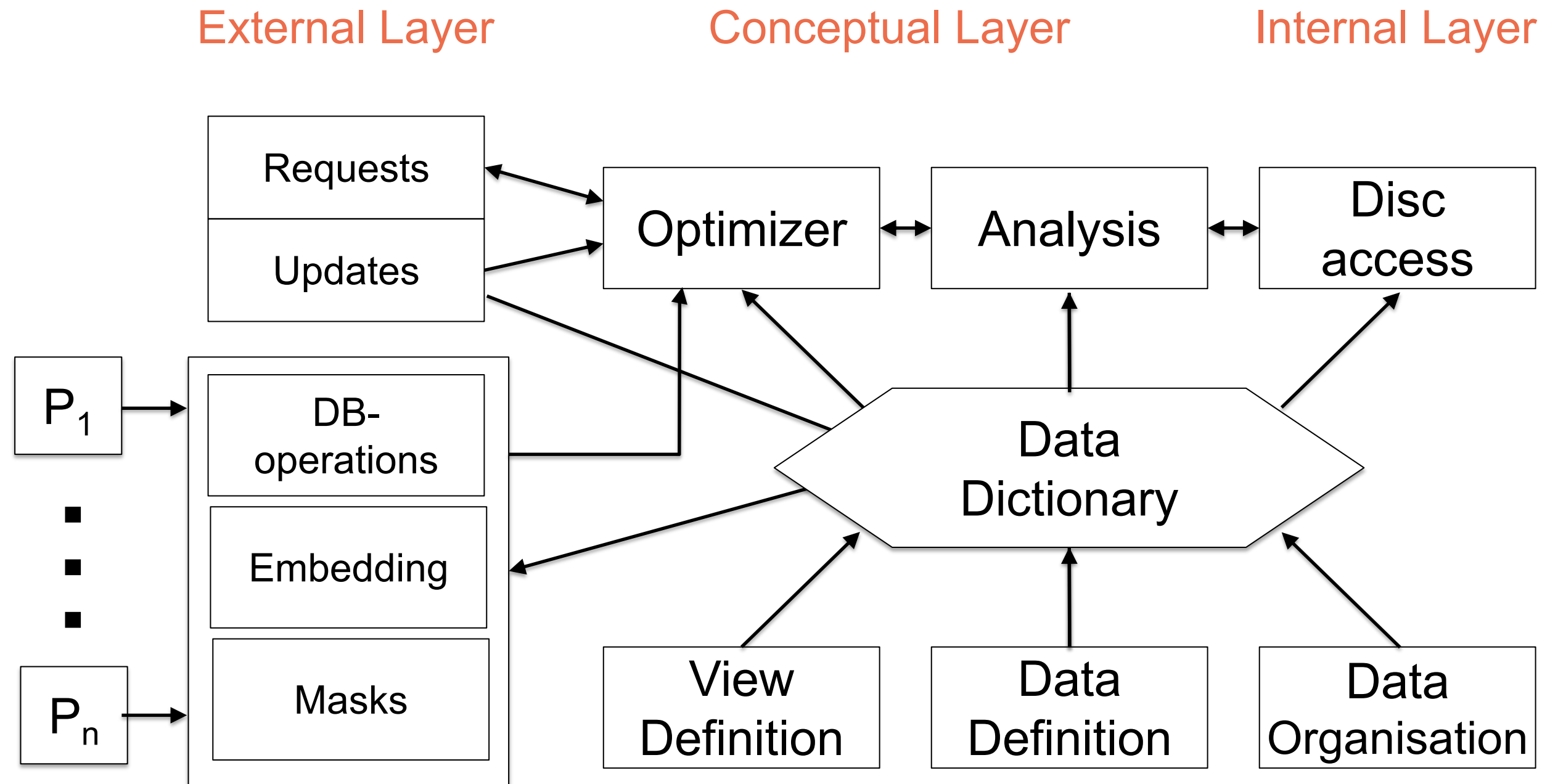
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Database System architecture – Function Modules

ANSI-SPARC Architecture

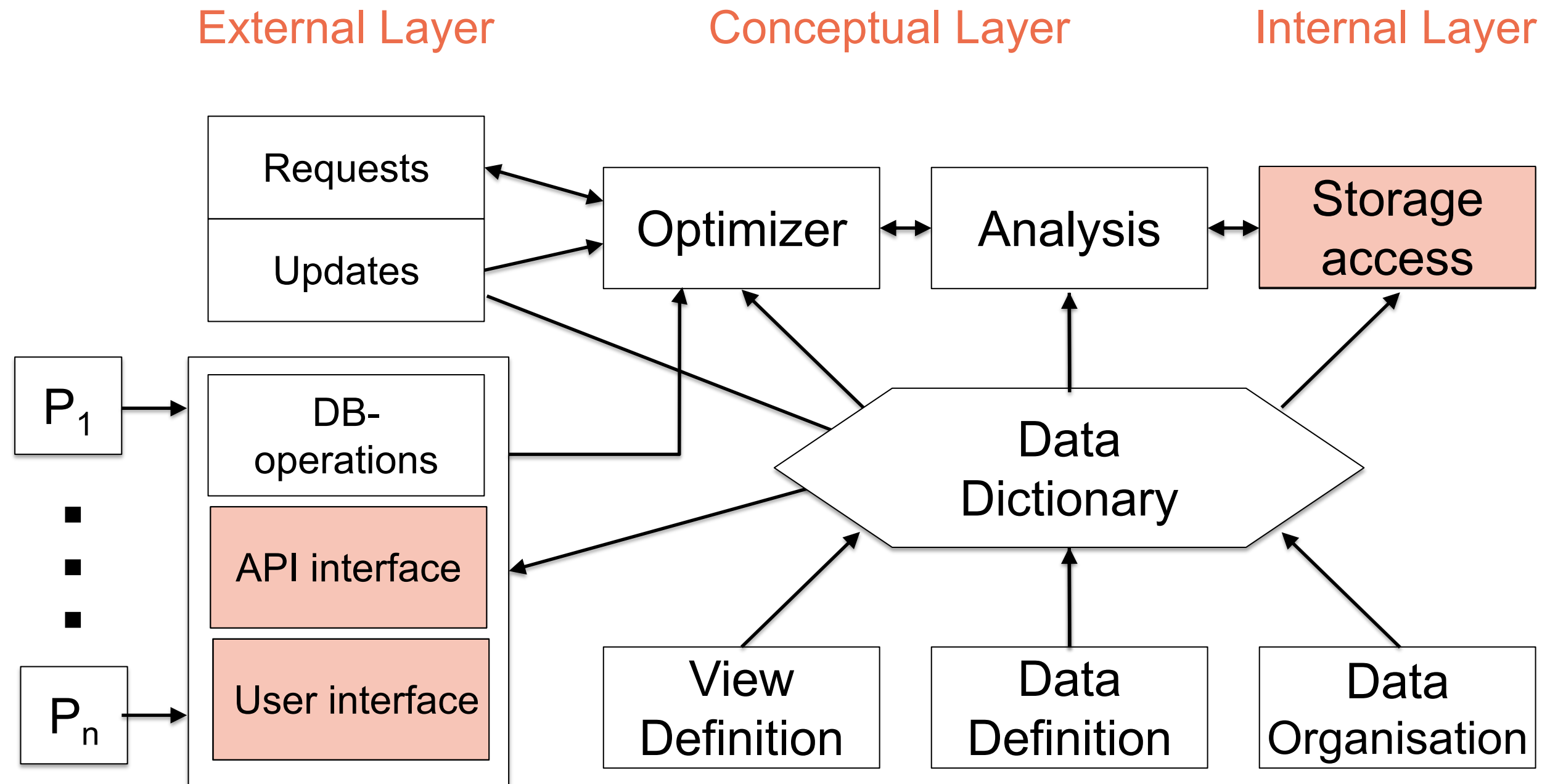
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Database System architecture – Function Modules

ANSI-SPARC Architecture – Modern Terms

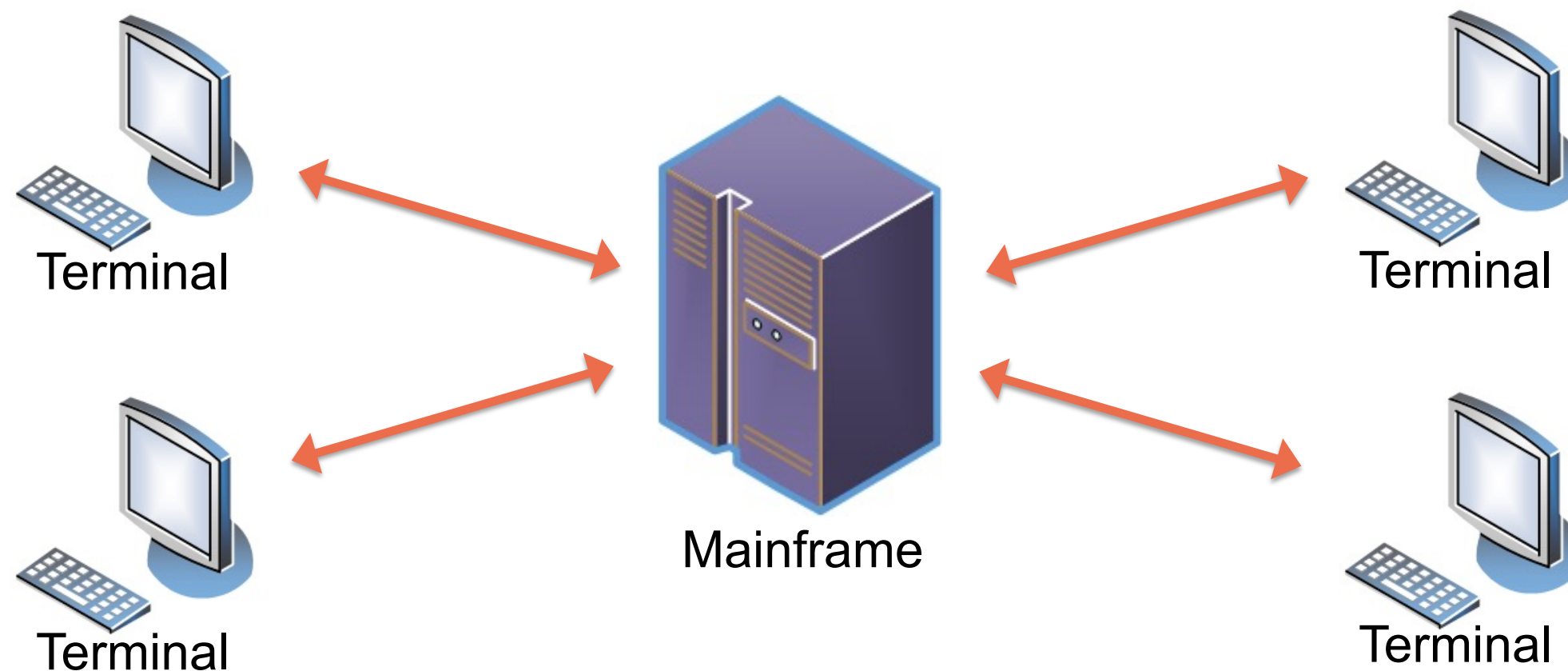
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Centralized DBMSs Architecture

Mainframe Systems

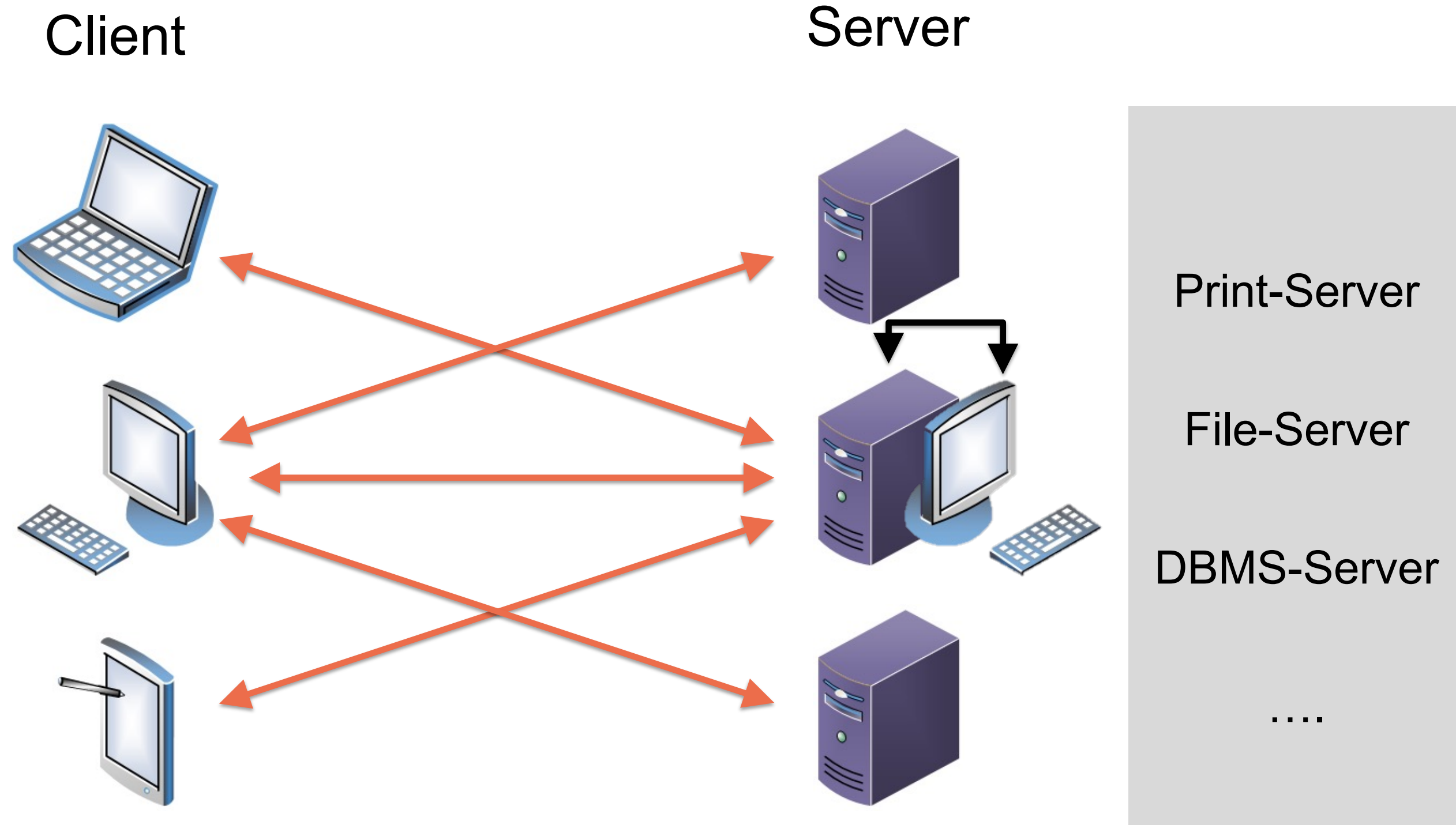
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- First professional database systems used mainframe computers and terminals with no processing power and only display capabilities
- The complete processing including application programs and user interface is provided by the mainframe computer
- This centralized approach demands immense power from the mainframe and is relatively inflexible.

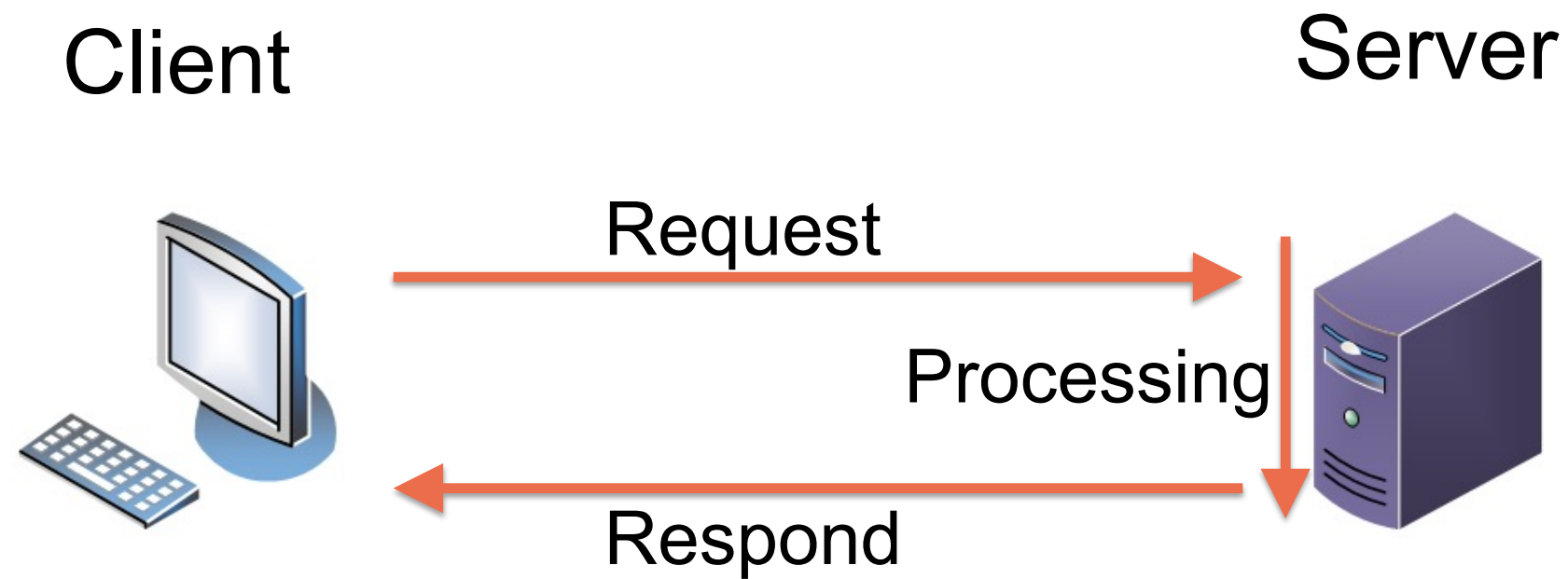
Client-Server Architecture

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Client-Server Architecture

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Presentation and User interaction (GUI)

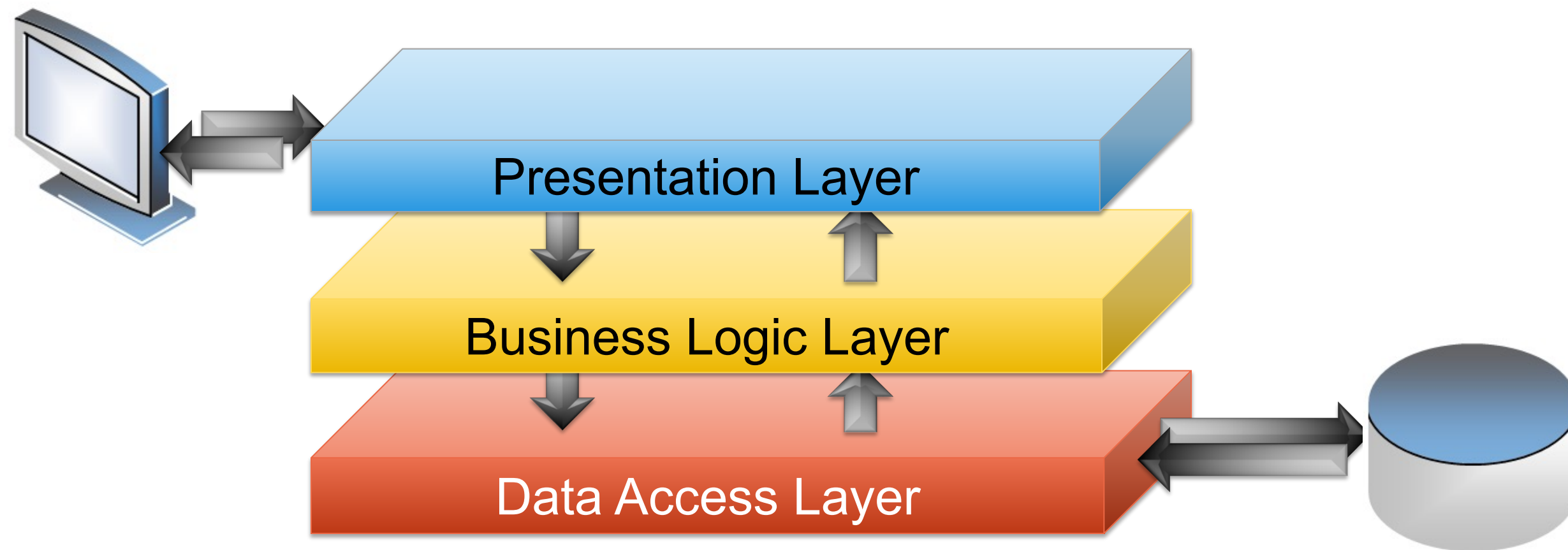
Application/Business Logic

Database Management System

Where are those functions?

Schematic Structure of a Database Application

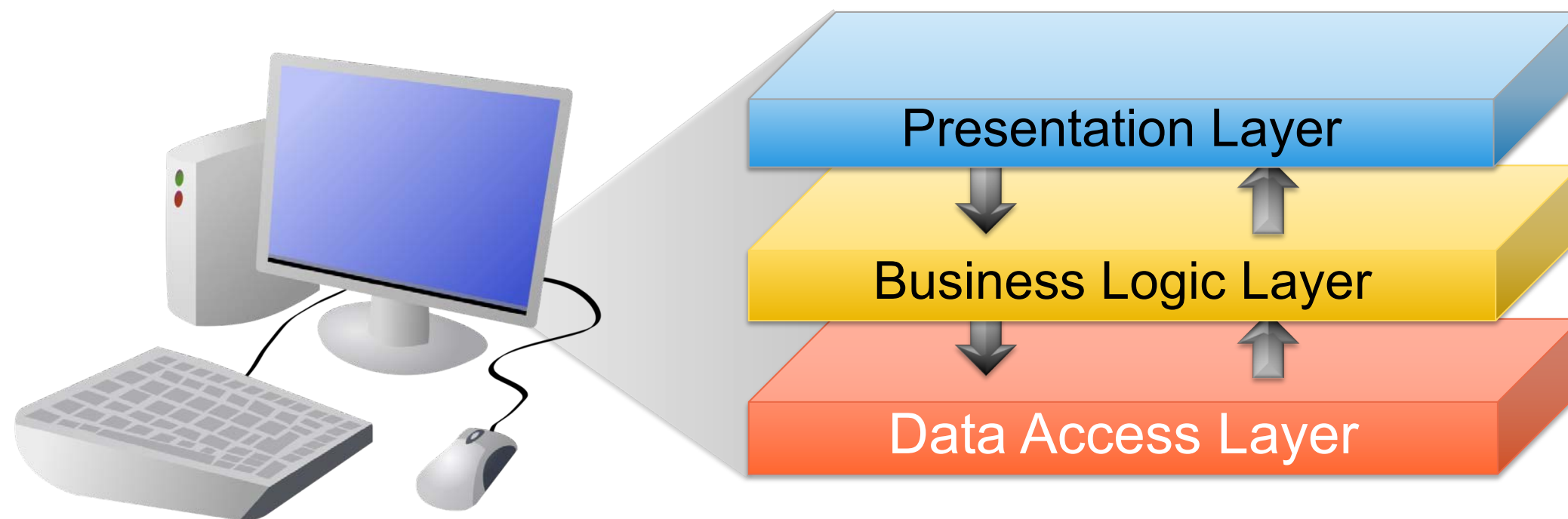
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Single Tier Applications

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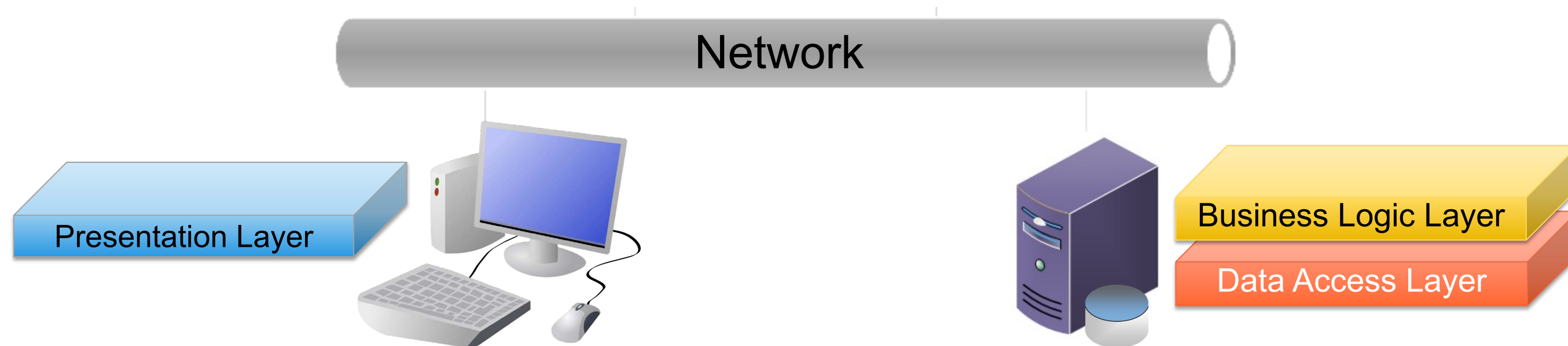
- All three layers are combined in one computer
- Typical: Desktop data base (e. g. SqlLite, Access desktop)
- For business demands usually not feasible (only for very specific tasks)



Two Tier Architecture – Intelligent Server

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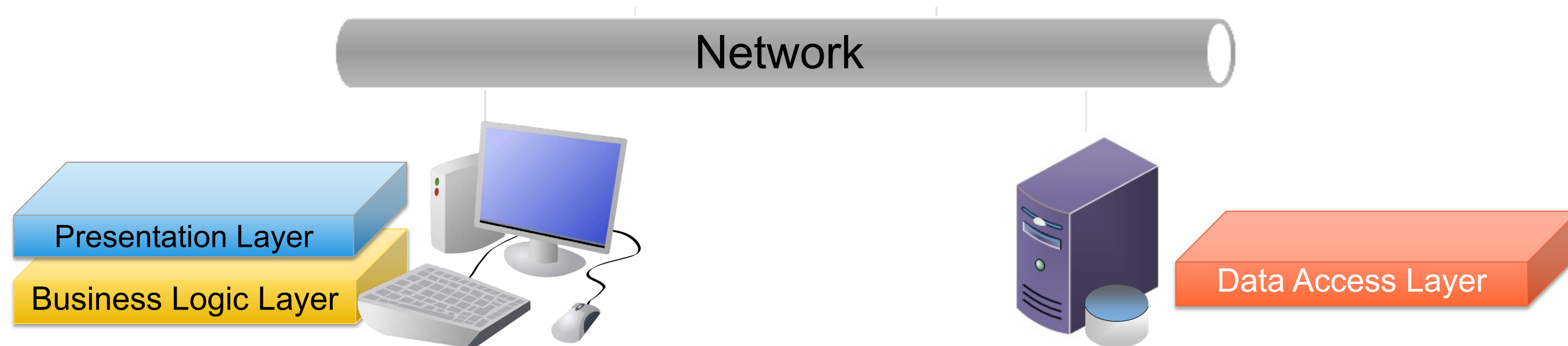
- The Two Tier Architecture is the typical Client Server Architecture
- Below the version with an "intelligent" server
- The advantage: even highly complex data manipulations can be provided though not every client could process them by its own
- The server needs sufficient resources to serve the maximum of estimated simultaneous clients



Two Tier Architecture – Intelligent Client

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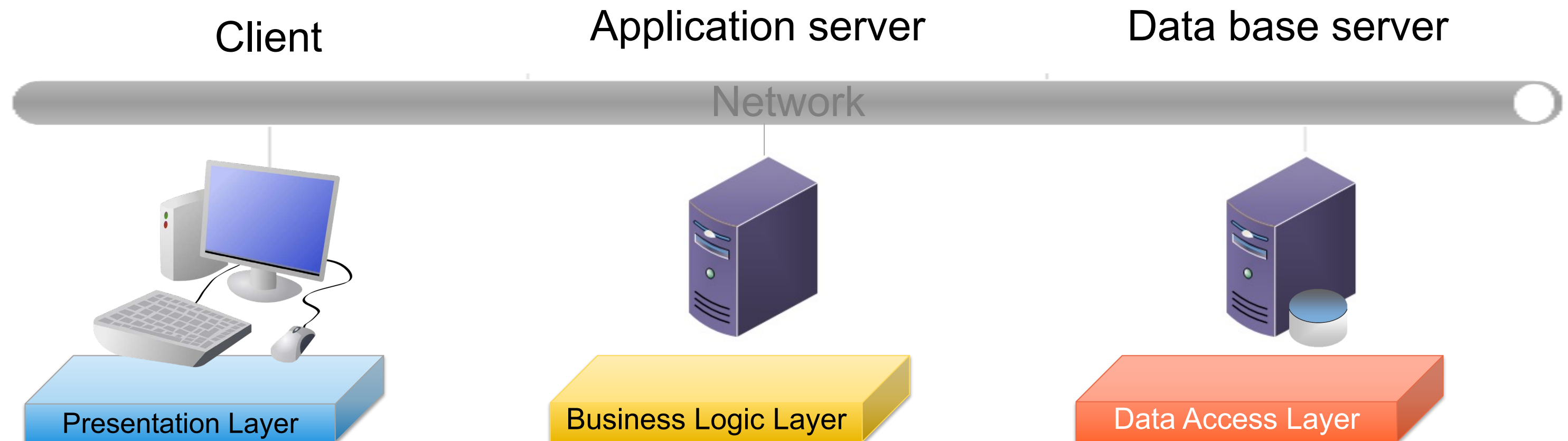
- The intelligent client is also known as Fat Client
- The advantage: the data needs to be transferred to the client only once
- Implementing the client software can prove problematic and the client devices often prove lacking when processes get complex
- In practise, often in between solutions are implemented – combining the advantages of both systems in regard of the task at hand. Data intense operations are executed on the server, the rest on the client



N Tier Architecture

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- The typical version is the three tier architecture
- This system is highly flexible as application and data base servers can be added as need be
- The client is typically thin in such a structure

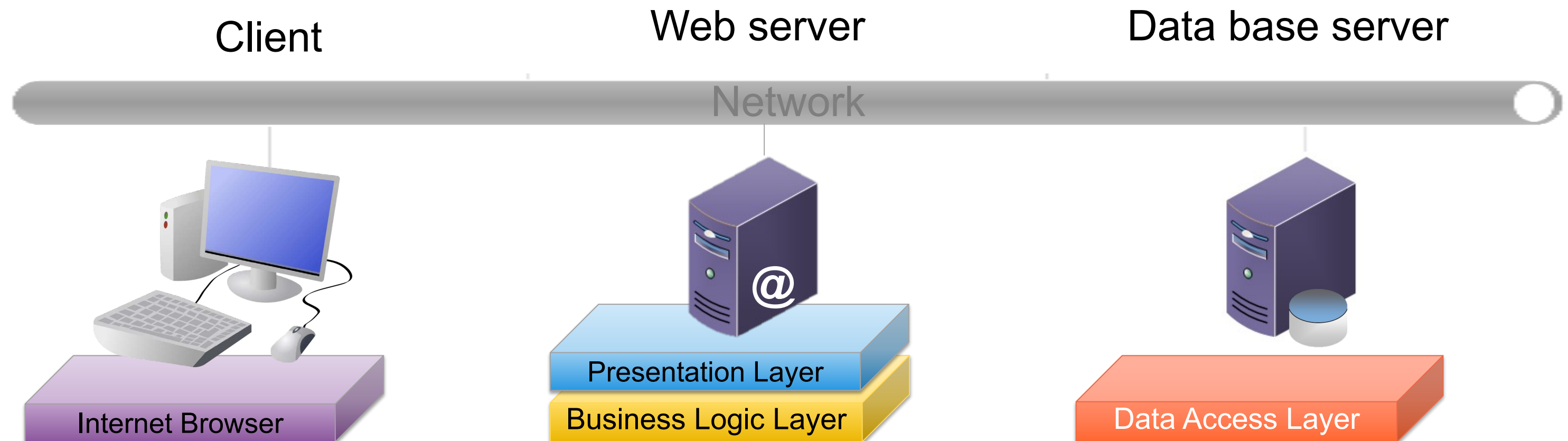


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N Tier Architecture – Internet Solution

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- The internet solution does not need any special software on the client, because the webserver delivers readymade data, every browser can display
- This is not only relevant regarding external users but also in the company, because installing and updating client software can be problematic in highly regulated company infrastructures



[Geis11] P. 345 [Kofl22] P.48

Architecture of a DBMS – Example

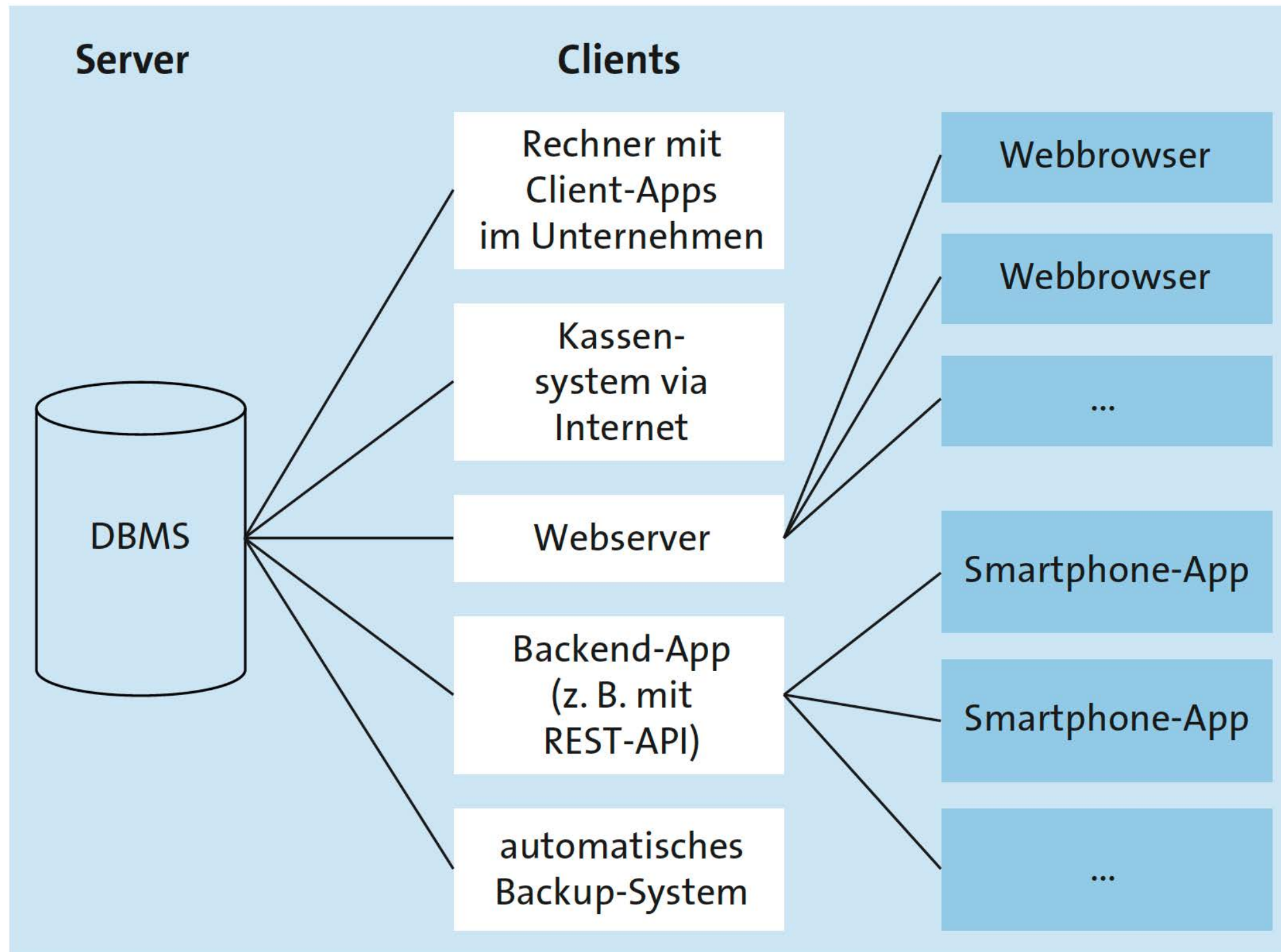
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A database server for is responsible for the cable car and lift operation in a ski area.

- The employees maintain the system and set the prices.
- Automated cash register systems for the purchase of ski tickets.
- A website provides information about the operation and the current capacity utilisation.
- A smartphone app documents the distance covered by skiers.
- An automatic backup is carried out every night.

Architecture of a DBMS – Example

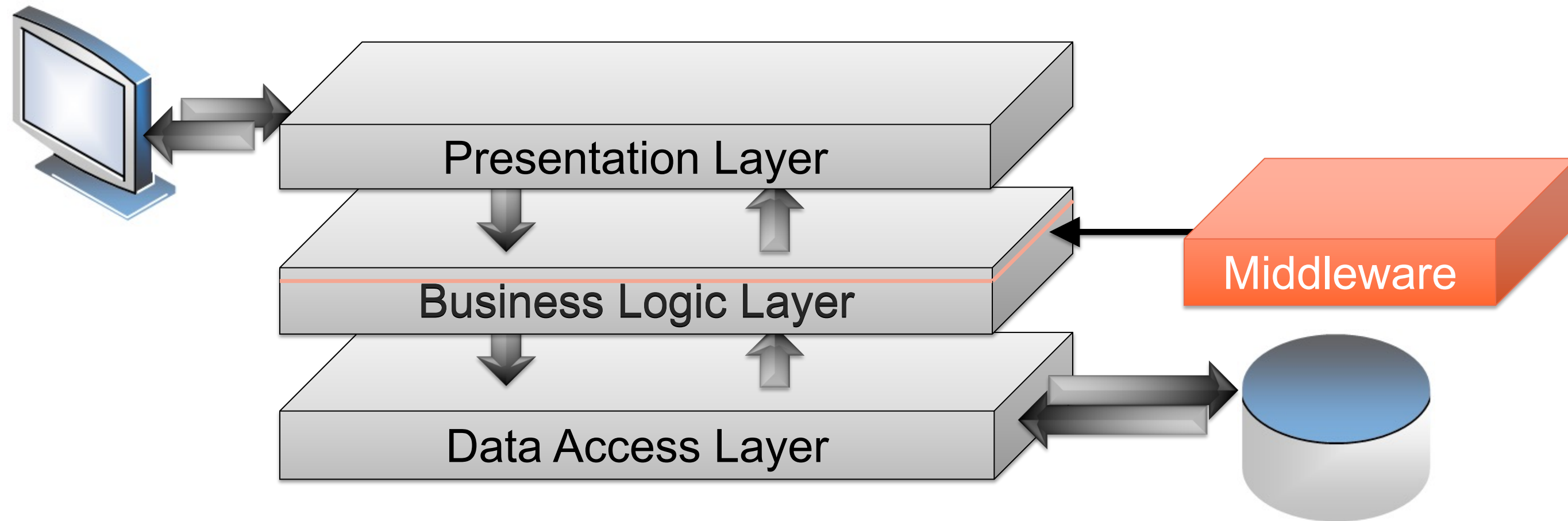
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[Kofl22:49]

Middleware – Translation Layer

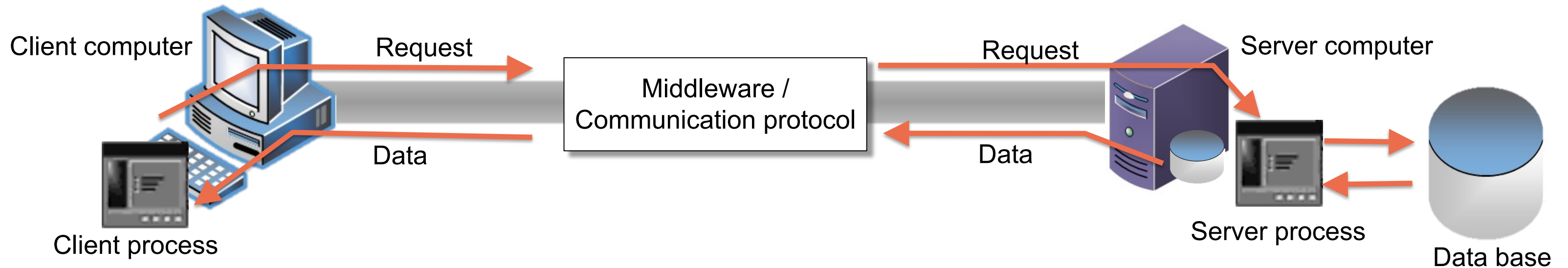
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- The communication between the different layers needs an interface as the layers are provided through different systems. This is also true within a layer if the services change.
- Middleware provides an abstract layer that translates commands and data between the different software of the system
- Changes in the databases are possible without afflicting the application if middleware accounts for the interaction

Client Server Communication – Middleware

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- Middleware is a service application, that creates an interface between server and client
- Middleware abstracts the database and network access of database applications. Thus the application does just access the middleware and does not have to take care of the communication to the (different) database or application server(s).

Client Server Communication – Middleware

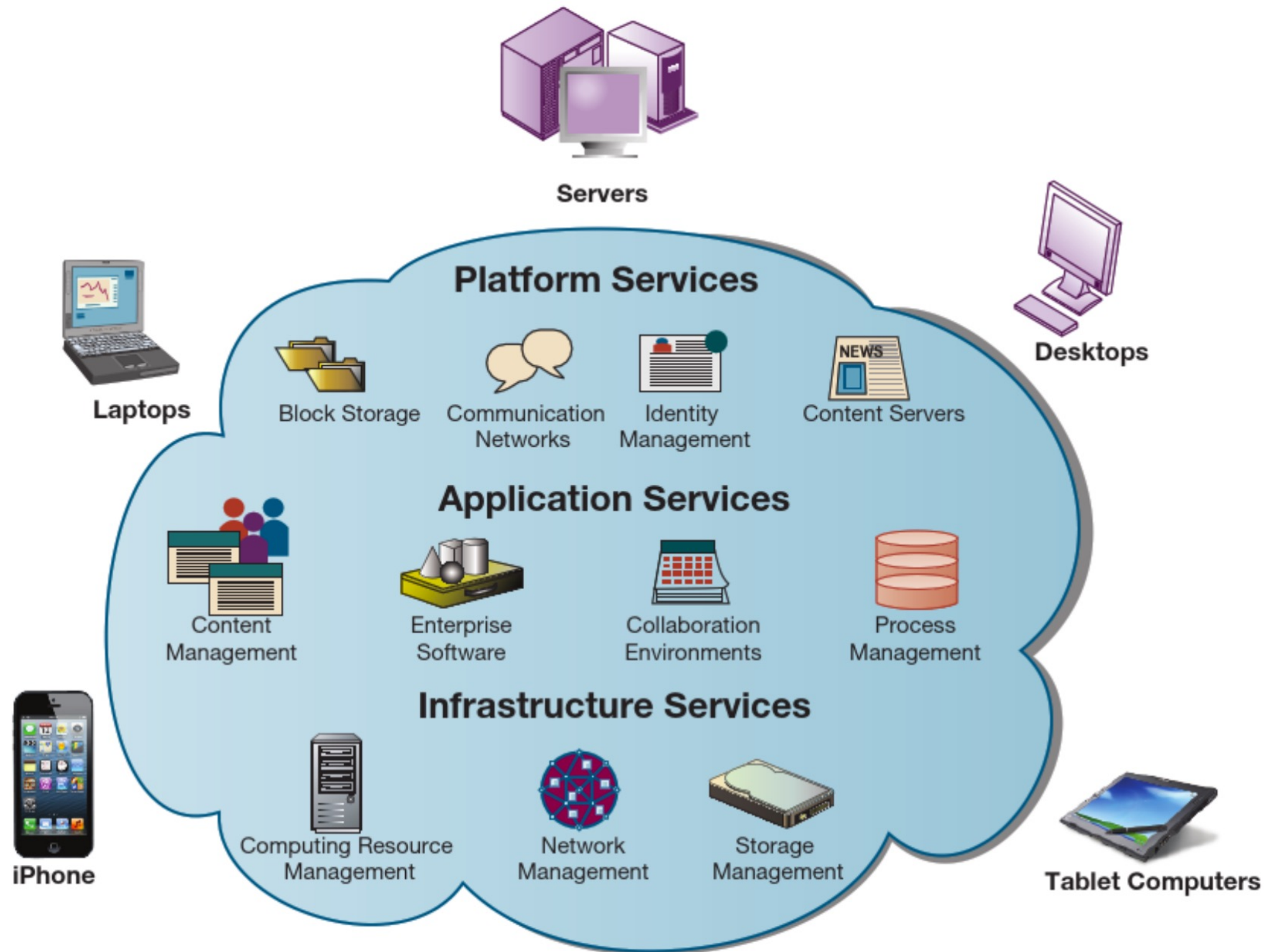
Web services

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- Web services provide the interfaces as service in the internet
- The web service provides web documents that are readable by the adjacent systems (e.g. JSON, XML)
- Today, XML is the most used exchange format with SOAP, AJAX or REST as protocols
 - JSON: JavaScript Object Notation
 - XML: Extensible Markup Language
 - SOAP: Simple Object Access Protocol
 - AJAX: Asynchronous JavaScript And XML
 - REST: Representational state transfer

Cloud solutions

20 November 2023



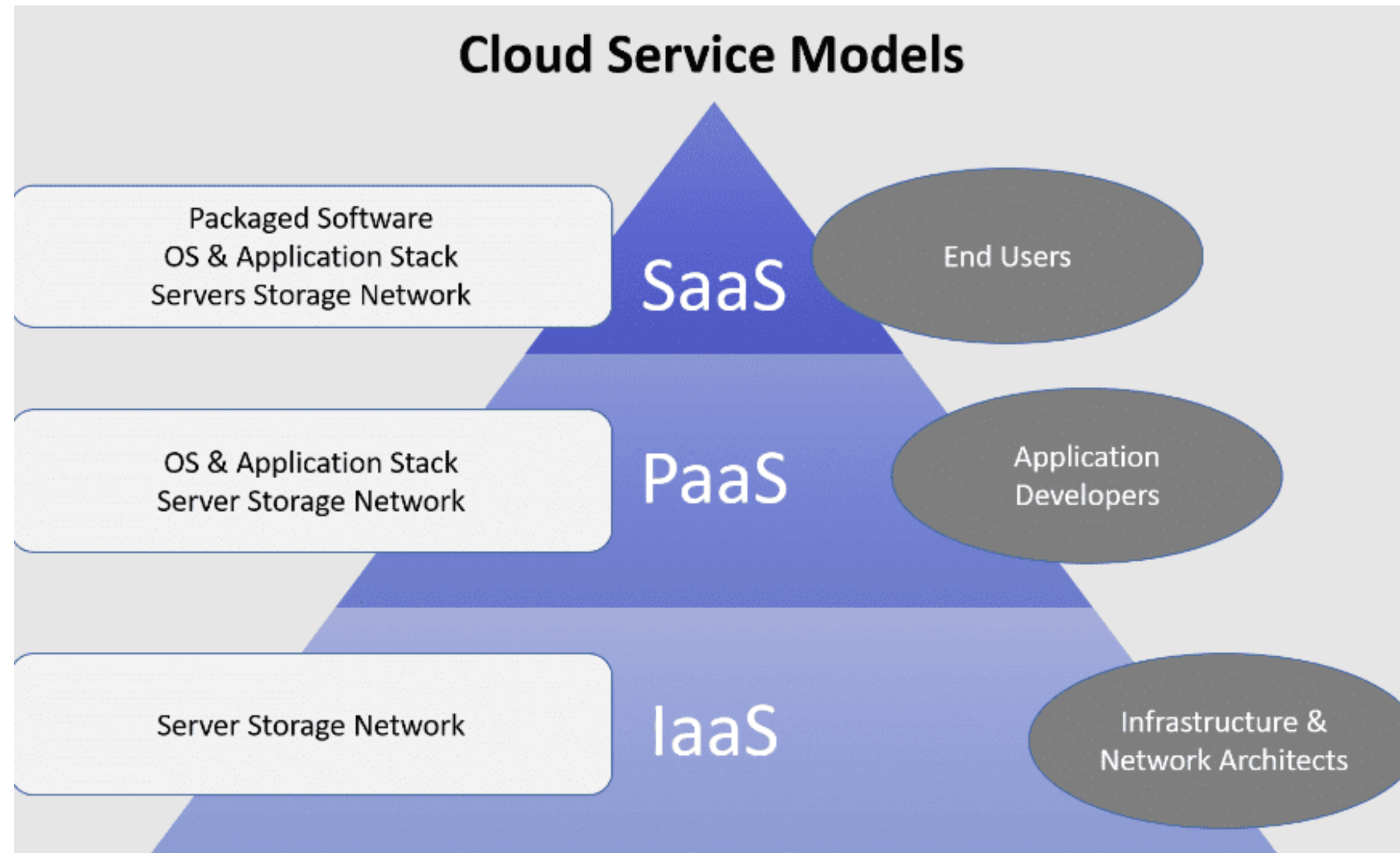
Cloud solutions

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- Cloud computing is a model of computing in which computer processing, storage, software, and other services are provided as a shared pool of virtualized resources over a network, primarily the Internet.
- **On-demand self-service**: Consumers can obtain computing capabilities such as server time or network storage as needed automatically on their own.
- **Ubiquitous network access**: Cloud resources can be accessed using standard network and Internet devices, including mobile platforms.
- **Location-independent resource pooling**: Computing resources are pooled to serve multiple users, with different virtual resources dynamically assigned according to user demand.
- **Rapid elasticity**: Computing resources can be rapidly provisioned, increased, or decreased to meet changing user demand.
- **Measured service**: Charges for cloud resources are based on the amount of resources actually used.

Cloud computing – service types

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Cloud computing – service types

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- **Software as a service (SaaS):** Customers use software hosted by the vendor on the vendor's cloud infrastructure and delivered as a service over a network.
 - Example: Google G Suite
- **Platform as a service (PaaS):** Customers use infrastructure and programming tools supported by the cloud service provider to develop their own applications.
 - Example: Microsoft Azure
- **Infrastructure as a service (IaaS):** Customers use processing, storage, networking, and other computing resources from cloud service providers to run their information systems.
 - Example: Amazon Elastic Compute Cloud or DynamoDB

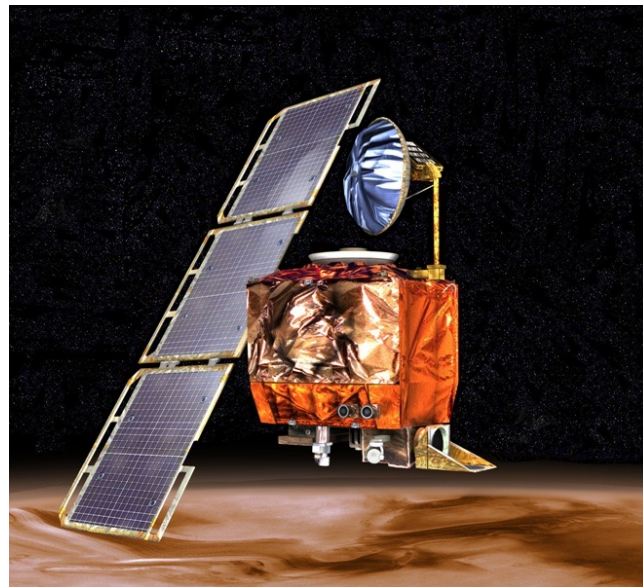
5. Architecture, Data Quality, Transactions

1. Basic System Structure
2. ANSI-SPARC Architecture
3. Data and Information Quality

Why Data Quality Management?

Examples what happens without

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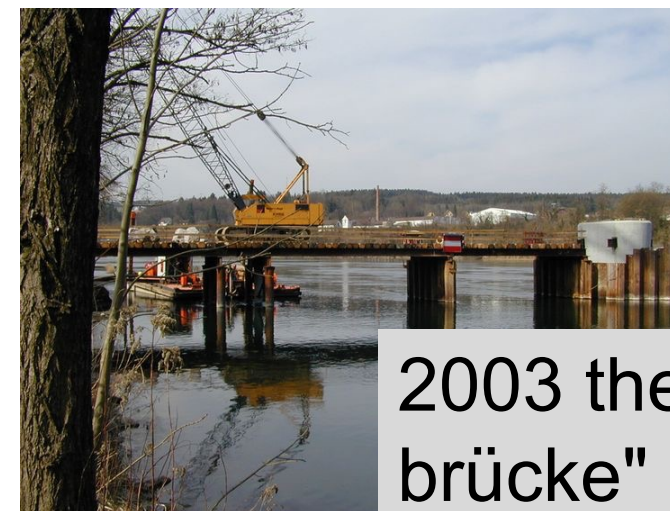


1999 NASA lost the Mars Climate Orbiter because of faulty translations of british units to the metric system



2005 a broker of Mizuho Fin tried to sell 610000 shares for 1 ¥ instead of 1 share for 610.000 ¥. It was an data input error

40% of all projects in companies today fail because of lacking data quality



2003 the "Hochrheinbrücke" at Laufenburg was built with 54 cm height difference because of a sign flaw

Bad Data Quality is Costly

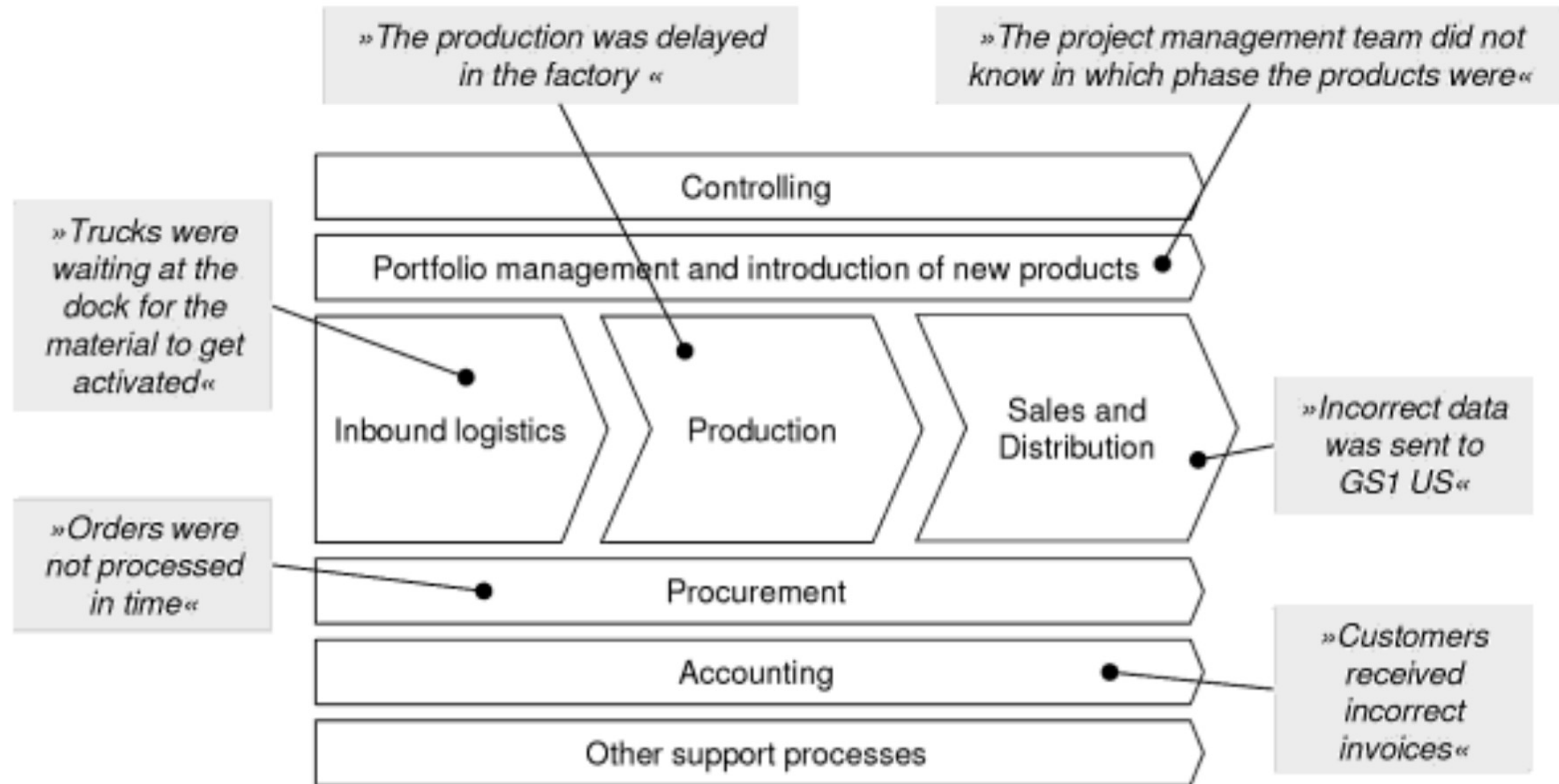
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- The average revenue lost by companies due to faulty data is up to **15 million dollars**.
In other words, the cost of poor data quality is **15% to 25% of revenue**
- **50%** of IT budgets are spent on data reprocessing
- Once a data series is ingested, it costs \$1 to verify it, \$10 to clean it, and.
\$100 if it remains erroneous

- Today data are a strategic resource for businesses.
- Data must be managed with regard to **time**, **cost** and **quality**
- Data Quality Management is a **business function** to optimize and maintain the data quality of the company
- **Data Governance** is the framework for DQM and defines which roles with which responsibilities assume the tasks of DQM.

Data Quality – why is it important?

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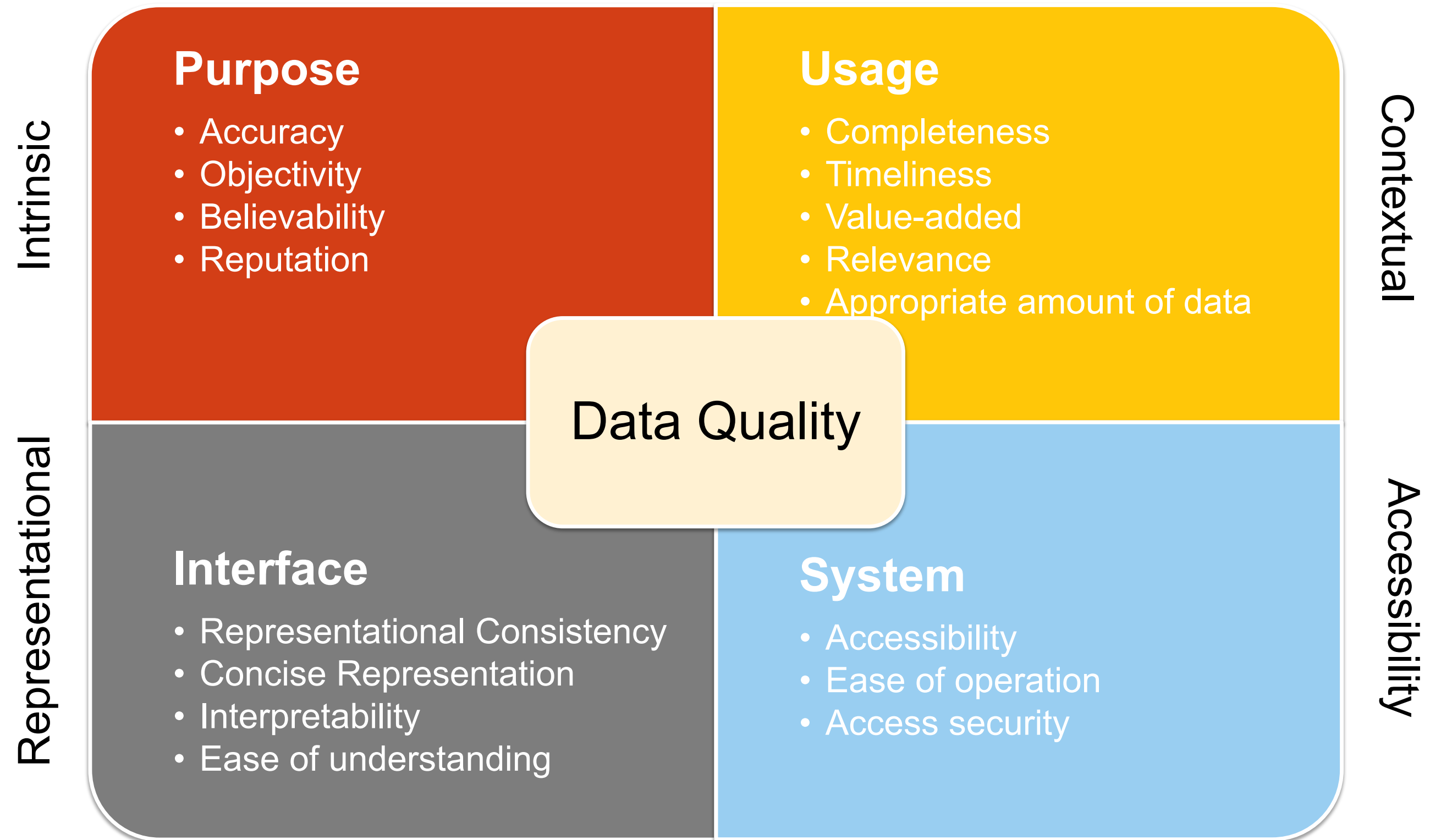
Data Quality – why is it important?

20 November 2023

1. **Accurate decision making:** Bad data skews the information source for evidence-based decision making
2. **Operational efficiency:** Business processes rely on data. Faulty data leads to inefficient and ineffective processes
3. **Compliance:** Conformance with Data compliance standards (such as GDPR, HIPAA, and CCPA) is only possible with clean and reliable data.
4. **Financial operations:** Operations such as making timely payments, preventing underpay / overpay incidents, etc are only possible with clean high-quality data.
5. **Customer personalization and loyalty:** Understanding the customer needs high-quality data
6. **Competitive advantage:** Knowing your market and trends ...
7. **Digitization:** Digitization can only be successful on the basis of reliable data

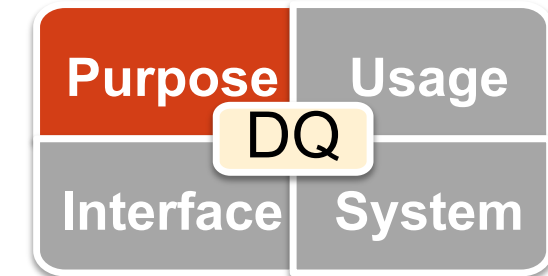
Dimensions of Data Quality

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Purpose – Intrinsic Data Quality

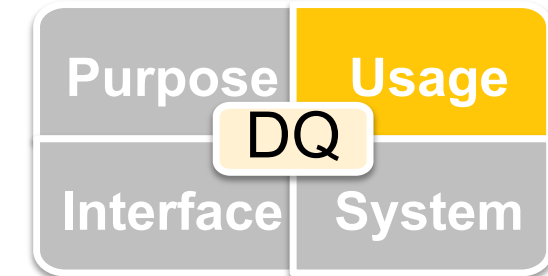
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- **Accuracy:** The extent to which data are correct, reliable, and certified free of error
- **Objectivity:** The extent to which data are unbiased (unprejudiced) and impartial
- **Believability:** The extent to which data are accepted or regarded as true, real, and credible
- **Reputation:** The extent to which data are trusted or highly regarded in terms of their source or content

Usage – Contextual Data Quality

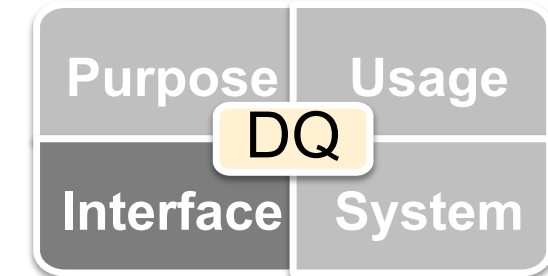
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- **Completeness:** The extent to which data are of sufficient breadth, depth, and scope for the task at hand.
- **Timeliness:** The extent to which the age of the data is appropriate for the task at hand.
- **Value-added:** The extent to which data are beneficial and provide advantages from their use.
- **Relevance:** The extent to which data are applicable and helpful for the task at hand.
- **Appropriate amount of data:** The extent to which the quantity or volume of available data is appropriate.

Interface – Representational Data Quality

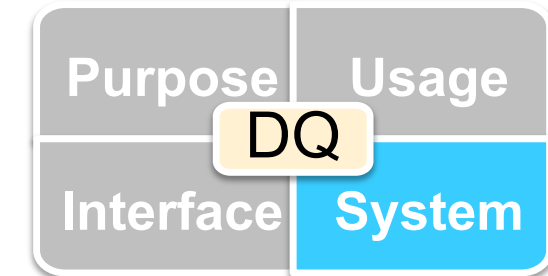
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- **Representational consistency:** The extent to which data are always presented in the same format and are compatible with previous data
- **Concise Representation:** The extent to which data are compactly represented without being overwhelming (i.e., brief in presentation, yet complete and to the point).
- **Interpretability:** The extent to which data are in appropriate language and units and the data definitions are clear
- **Ease of understanding:** The extent to which data are clear without ambiguity and easily comprehended

System – Accessibility Data Quality

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- **Accessibility:** The extent to which data are available or easily and quickly retrievable
- **Ease of operation:** The extent to which data are easily managed and manipulated (i.e., updated, moved, aggregated, reproduced, customized).
- **Access security:** The extent to which access to data can be restricted and hence kept secure.

First:

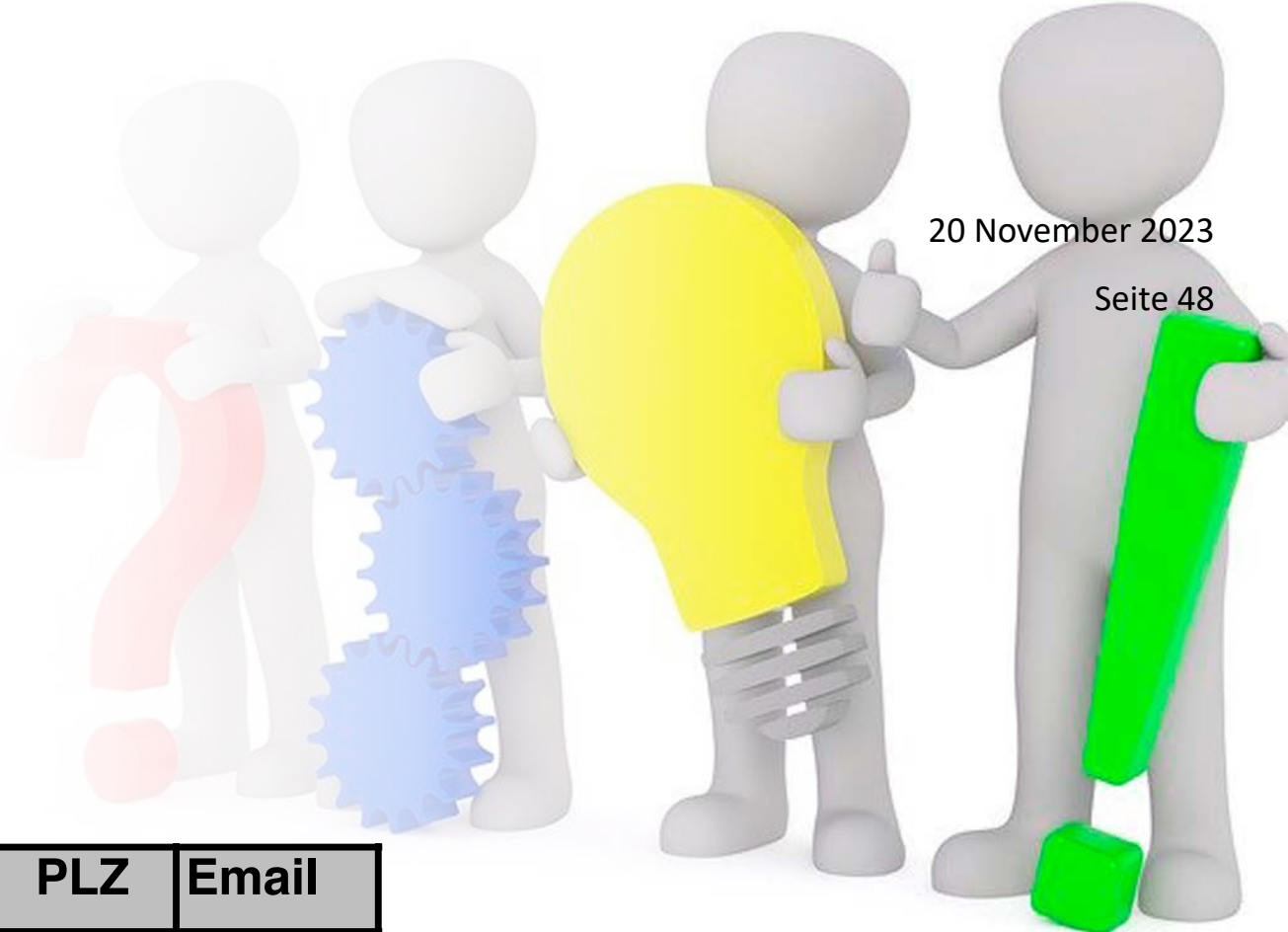
"How do I acquire the data?"

Then:

"How can I edit (select, aggregate, combine) the data, to get the needed information?"

Now:

"Which questions give me the best insights?"



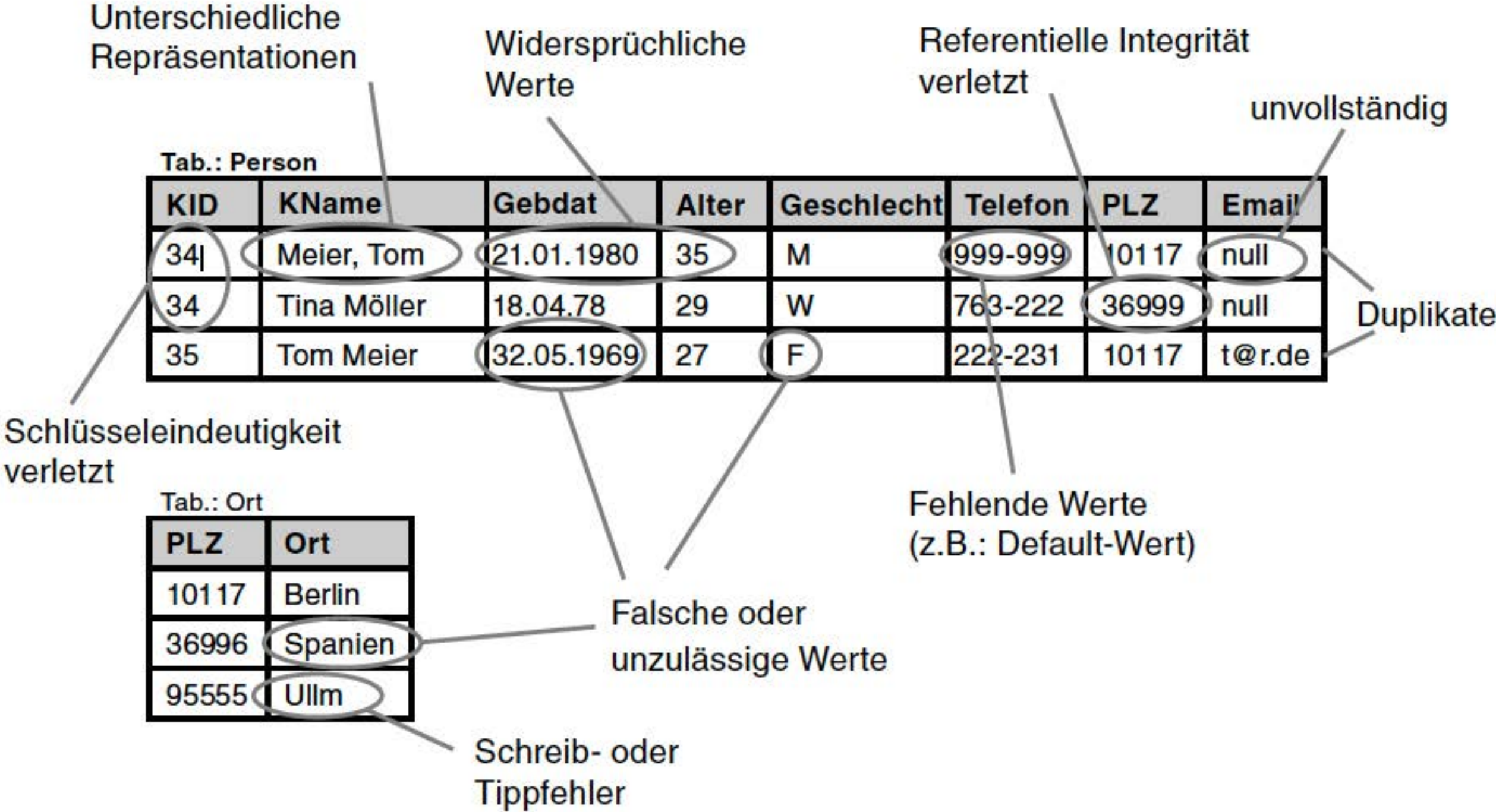
- Find the faulty data

Tab.: Person

KID	KName	Gebdat	Alter	Geschlecht	Telefon	PLZ	Email
34	Meier, Tom	21.01.1980	35	M	999-999	10117	null
34	Tina Möller	18.04.78	43	W	763-222	36999	null
35	Tom Meier	32.05.1969	52	F	222-231	10117	t@r.de

Tab.: Ort

PLZ	Ort
10117	Berlin
36996	Spanien
95555	Ullm



1. Initialisation:

Prepare and initiate the process before the start

- **Strategy** → Data quality must be part of the corporate strategy
- **Prioritising the goals** → Which data, system, application or process are the most important?
- Define the **roles and responsibilities** like data quality commissioner, data commissioner, data analyst

2. Define the data quality standards and requirements

- Judicial
- External and internal
- Technical
- Central provision of the standards and rules

Example of a Data Rule

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Field	Not Null	Distinct values	Absolute Distinctness	Relative Distinctness
SALUTATION	100.00%	42	0.45%	0.45%
HOME_PHONE_NUMBER	100.00%	175	1.88%	1.88%
LAST_NAME	100.00%	5,130	55.07%	55.07%
MIDDLE_NAME	100.00%	117	1.26%	1.26%
FIRST_NAME	100.00%	2,113	22.68%	22.68%
EMAIL_ADDRESS	100.00%	4,362	46.83%	46.83%

Contact must have valid salutation			
File			
Base Information			
Object Name			
Contact must have valid salutation			
Author		Creation Date	
DEFAULT_ADMIN		Jul 27, 2007	
Content Rule Information			
Data	Completeness	Format	Range
Plausibility	Accuracy	Generate	Errorlists
Description			
Admissible Values			
'Mr.', 'Ms.'			

SALUTATION	FREQ	PERCENT	PERCENT (A...
	3936	42.25443	42.25443
Mr.	3225	34.62158	76.87601
Ms.	1080	11.5942	88.47021
Mr	393	4.219	92.68921
Ms	226	2.42619	95.1154
Mrs.	162	1.73913	96.85453
Herr	98	1.05207	97.9066
Miss.	66	0.70853	98.61513
Dr.	35	0.37574	98.99087
Frau	18	0.19324	99.18411
Herr Dr.	12	0.12882	99.31293
Mrs	11	0.11809	99.43102
Ing.	5	0.05368	99.4847
Sr	4	0.04294	99.52764
Sr.	3	0.03221	99.55985

Finished. Total number of rows... Scroll to column:

Source: [Weig18:76]

Total Data Quality Management

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You can't control
what you can't
measure

2. Measuring of the existing data quality

The data is evaluated along the above mentioned
Dimensions of Data Quality

Before defined factors shows the measured quality

3. Analysis of the fault causes

- Where is the source: data, processes, or humans?
- Planning and design of the optimisation actions

4. Improvement of the data quality

- Data cleansing
- Software optimisation
- Business process optimisation
- Training of the employees

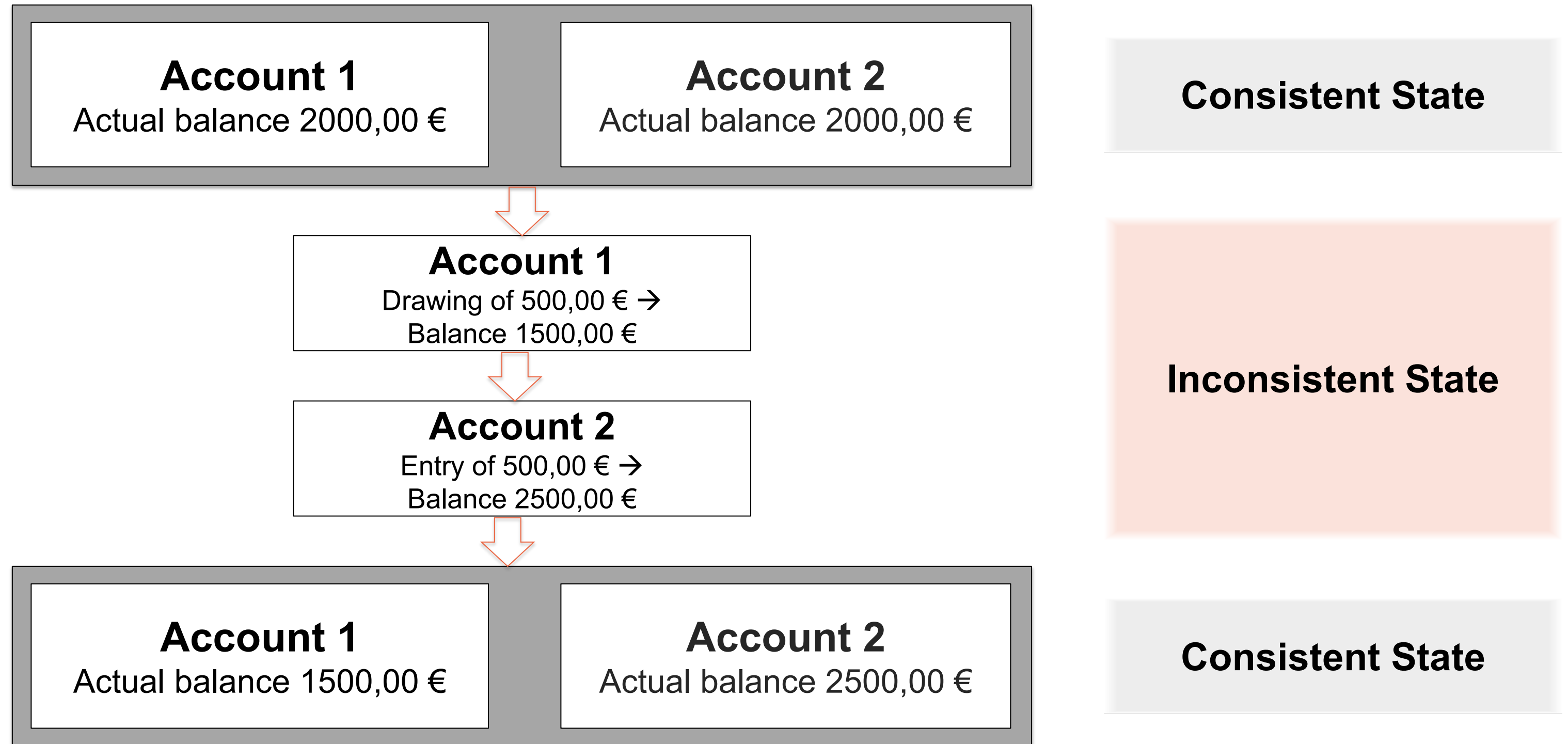
5. Permanent supervision of the data quality

5. Architecture, Data Quality, Transactions

1. Basic System Structure
2. ANSI-SPARC Architecture
3. Data Quality
4. Transaction

Transaction

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Causes of transaction failures

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- 1. A computer failure (system crash)**
hardware, software or network error during transaction execution
- 2. A transaction or system error**
a operation fails – e.g. through a integer overflow or a logical error
- 3. Local errors or exception conditions detected by the transaction**
e.g. data is missing or insufficient account balance for a withdrawal
- 4. Concurrency control enforcement**
transaction violating serialisation is terminated
- 5. Disk failure**
read or write malfunctions or a head crash
- 6. Physical problems and catastrophes**
fire, theft, sabotage ...

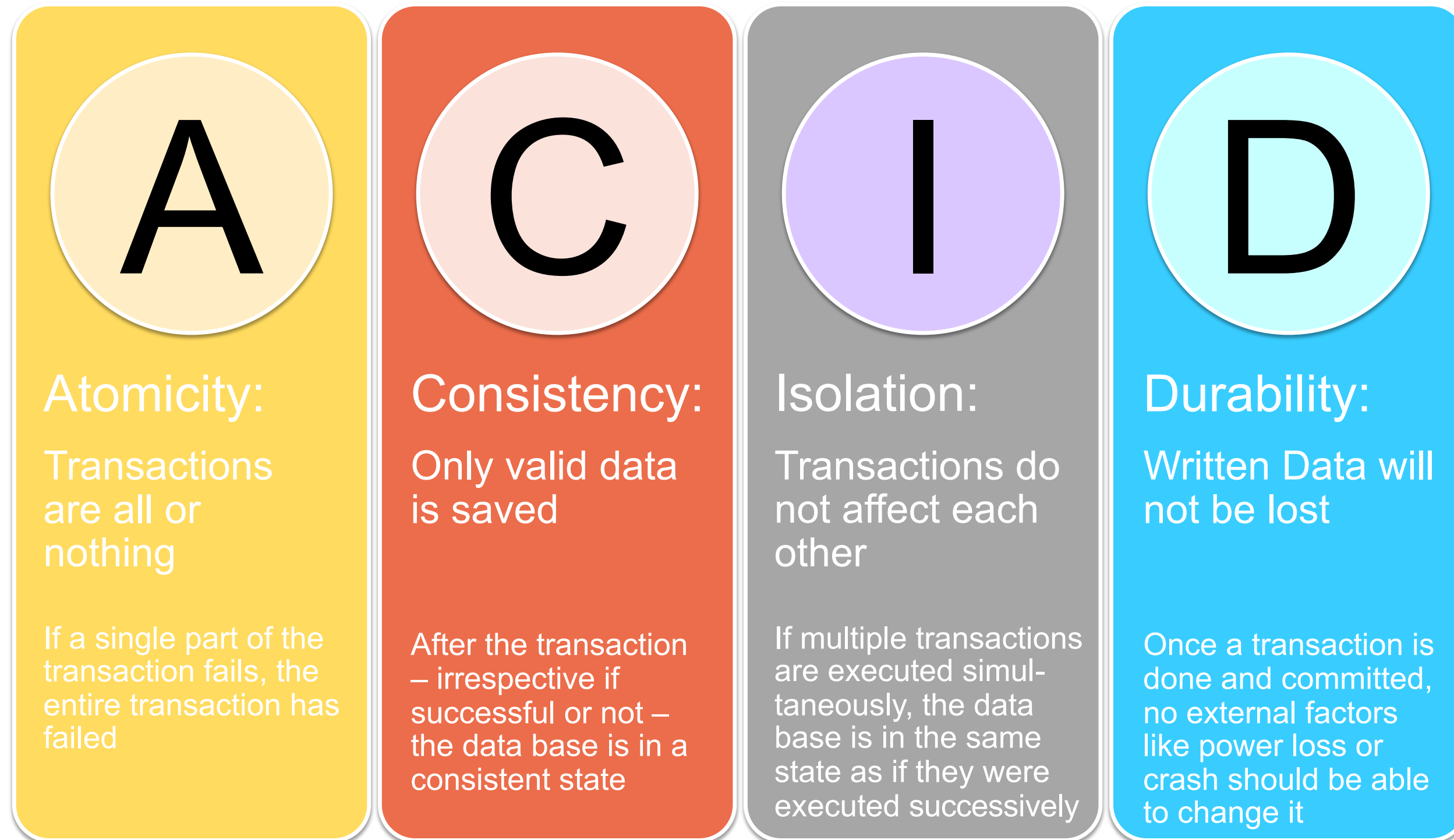
Example

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- Try this Step-by-step on your computer
 1. `SELECT * FROM Course WHERE courseNo = 'IS480';`
 2. `BEGIN TRY`
`INSERT INTO`
`Course(courseNo, crsDesc, crsUnits)`
`VALUES ('IS480', 'Transactions', 4);`
`END TRY`
`BEGIN CATCH`
`END CATCH;`
 3. `SELECT * FROM Course WHERE courseNo = 'IS480';`
 4. `UPDATE Course SET crsUnits= 7`
`WHERE courseNo = 'IS480';`
 5. `SELECT * FROM Course WHERE courseNo = 'IS480';`

Transactions – mandatory properties

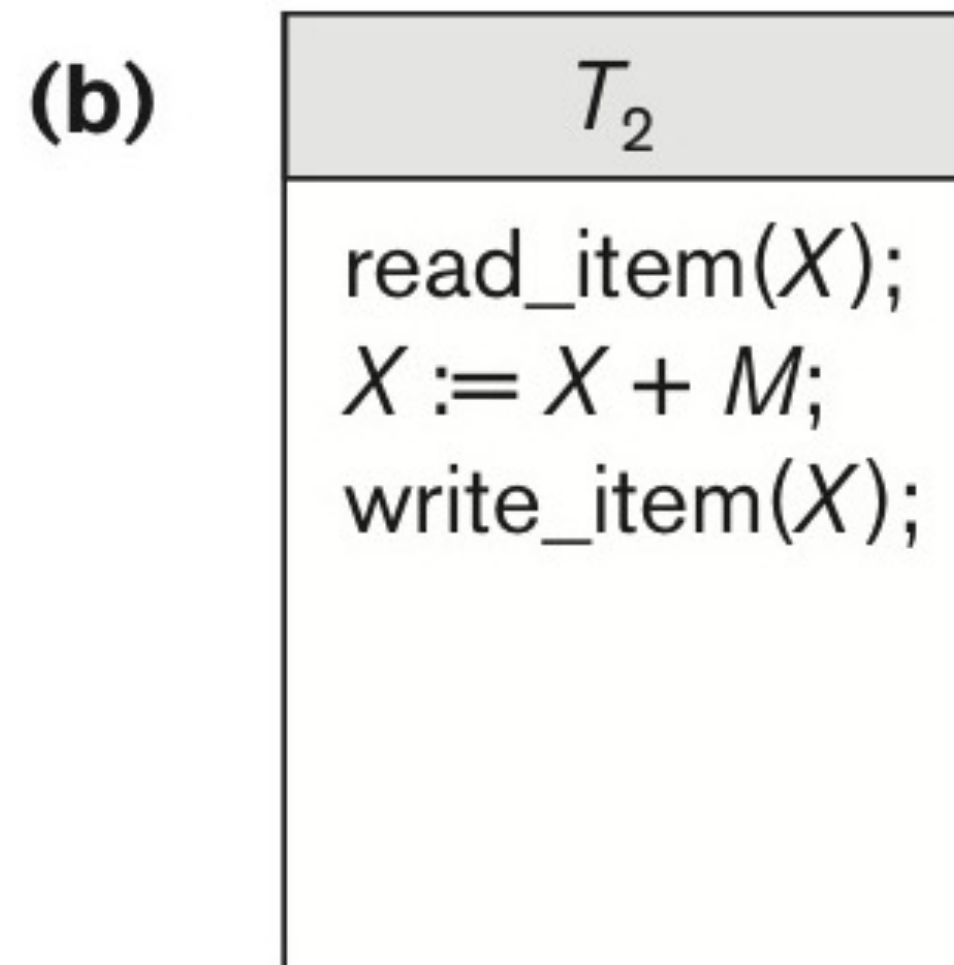
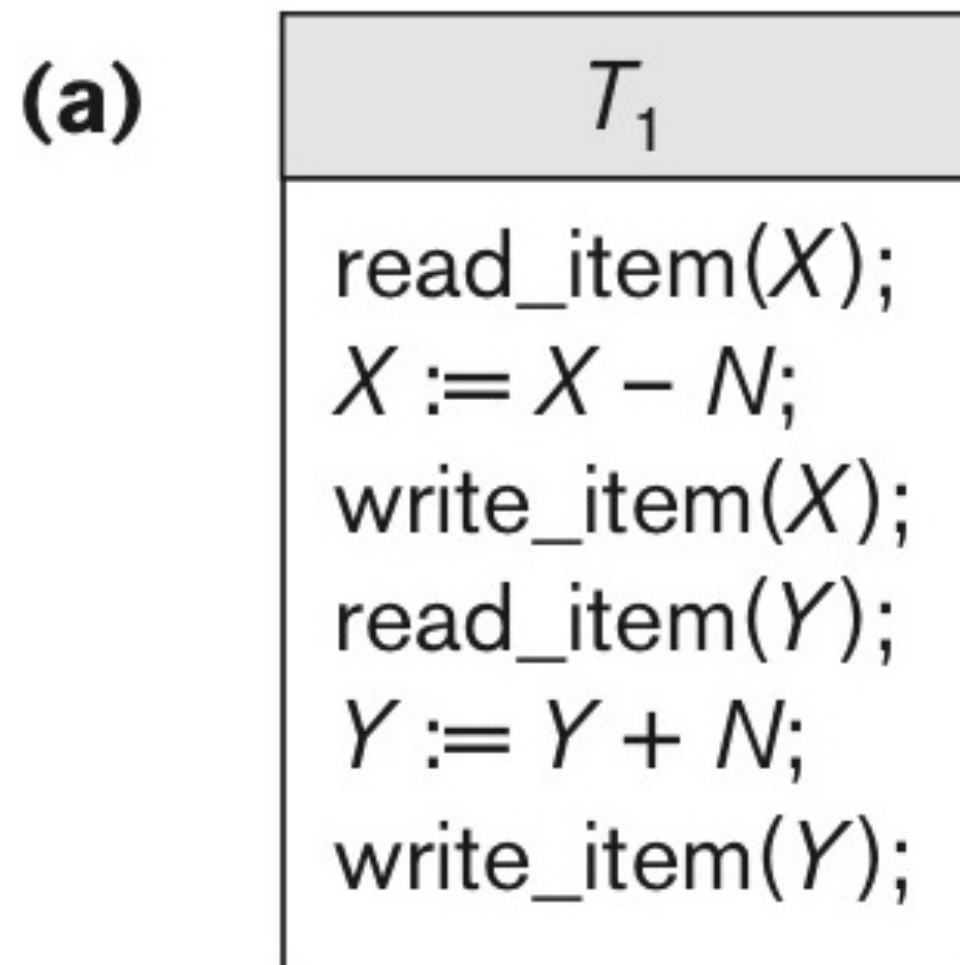
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Concurrency control

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- Two sample transactions. (a) Transaction **T1**. (b) Transaction **T2**.
(a) – Transferring N from X to Y
(b) – Adding M to X

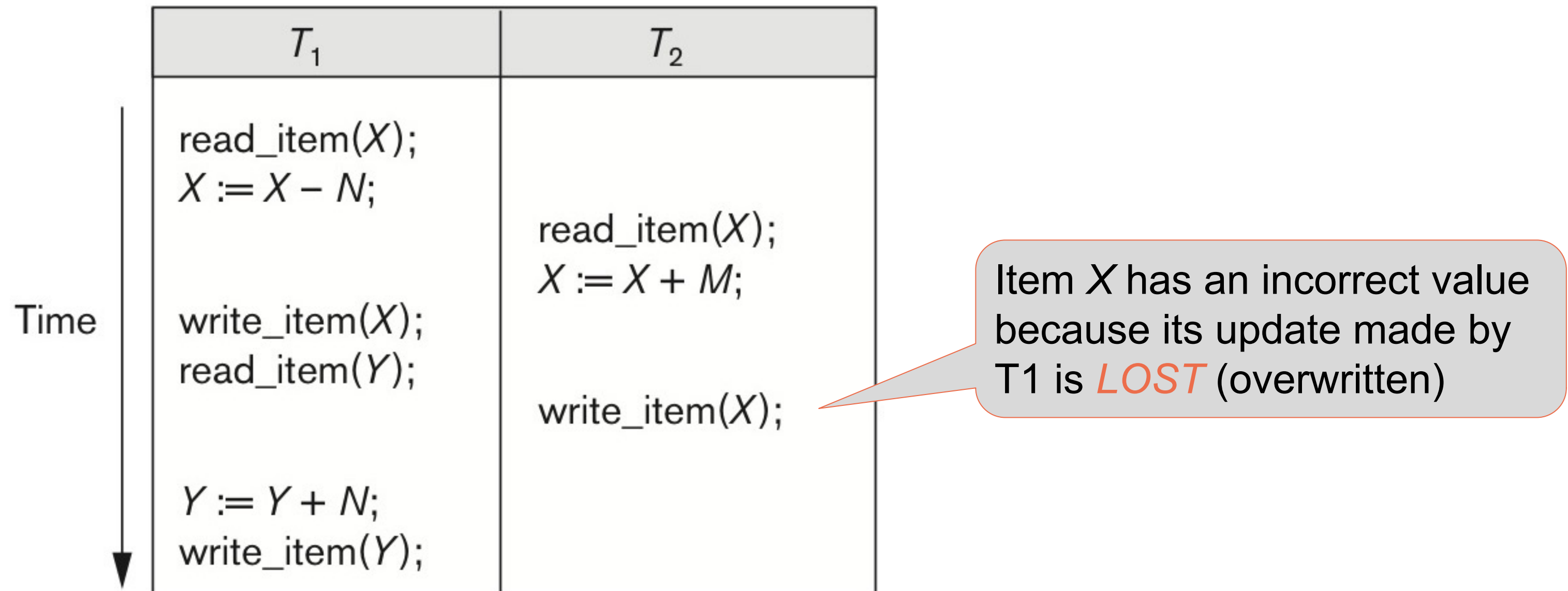


Concurrency control

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- Some problems that occur when concurrent execution is uncontrolled

The lost update problem

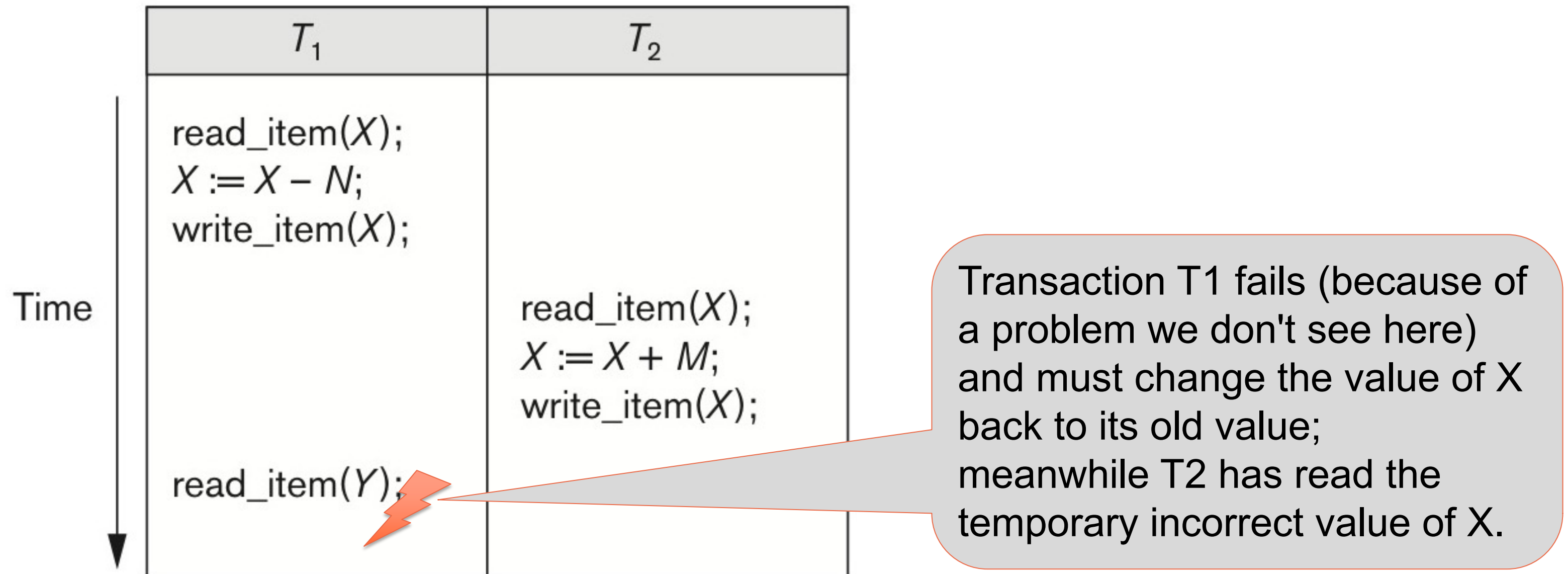


Concurrency control

20 November 2023

- Some problems that occur when concurrent execution is uncontrolled

The temporary update problem (dirty read)

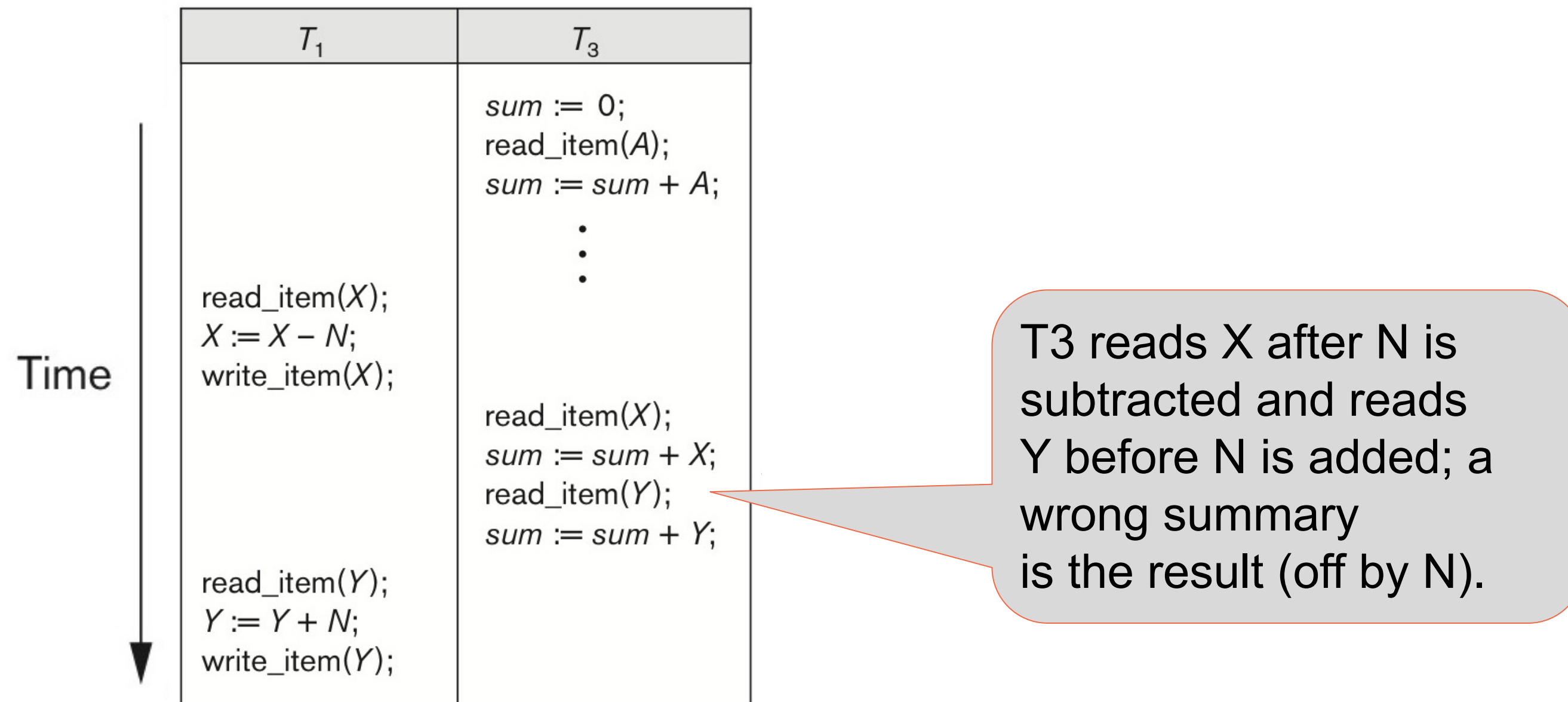


Concurrency control

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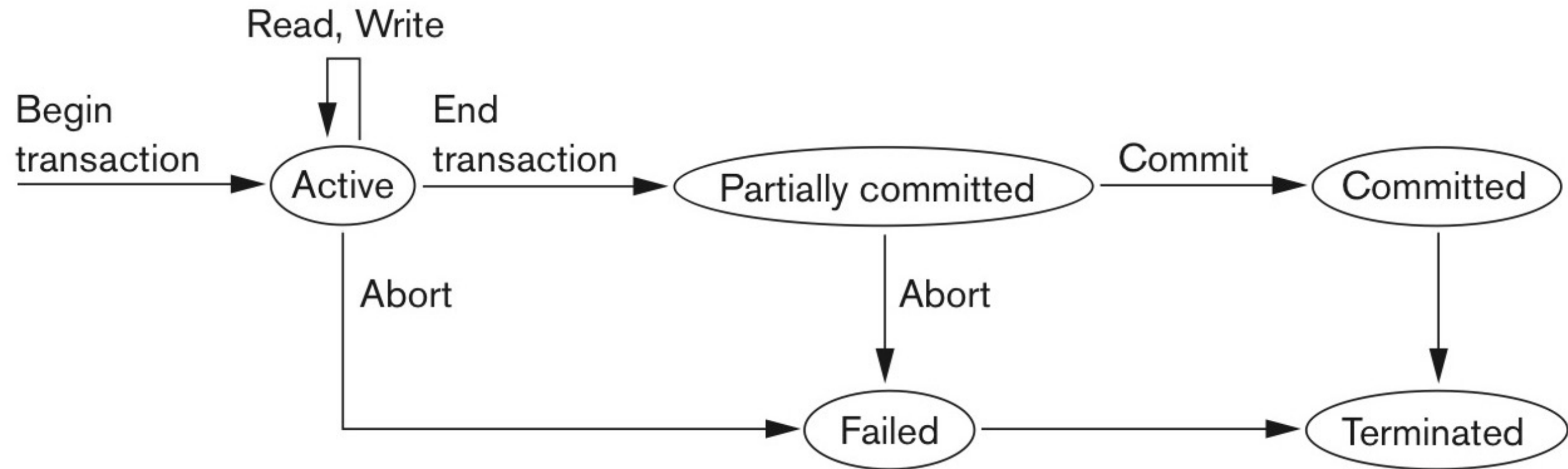
- Some problems that occur when concurrent execution is uncontrolled

The incorrect summary problem



Transaction states

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- BEGIN_TRANSACTION
- READ or WRITE
- END_TRANSACTION
- COMMIT_TRANSACTION
- ROLLBACK (or ABORT)

Try in SQL

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1. Try:

```
BEGIN TRANSACTION
```

```
UPDATE Course SET crsUnits= 15
```

```
WHERE courseNo = 'IS480';
```

```
SELECT * FROM Course WHERE courseNo = 'IS480';
```

```
ROLLBACK TRANSACTION
```

```
SELECT * FROM Course WHERE courseNo = 'IS480';
```

2. Change ROLLBACK to COMMIT

Try again

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